

Cross-Cohort Generalization Failure in Gut Microbiome-Based Alzheimer's Disease Classifiers: Evidence from Five Independent Cohorts with Batch Correction Analysis

Running title: Microbiome AD classifier generalization failure

Authors: Arhan Chakravarthy¹

Affiliations: ¹ Independent Researcher, Cupertino, CA

Correspondence: arhanc360@gmail.com

Keywords: gut microbiome; Alzheimer's disease; machine learning; cross-cohort generalization; batch correction; SHAP; PERMANOVA; 16S rRNA sequencing

Abstract

Gut microbiome dysbiosis has been proposed as a biomarker for Alzheimer's disease (AD) and mild cognitive impairment (MCI), yet individual studies consistently report contradictory taxonomic signatures across populations, and cross-cohort classifier performance has received limited systematic evaluation. We assembled five publicly available 16S rRNA gene sequencing cohorts (509 samples; China, Kazakhstan, South Korea; Zhuang 2018, Ling 2020, Zhu 2022, Kaiyrykyzy 2022, Kim 2022) and evaluated whether machine learning classifiers trained within one cohort could generalize to independent cohorts, whether standard batch correction methods could recover transferability, and which taxonomic features drove cohort-specific predictions. DADA2 amplicon sequence variant (ASV) processing and genus-level harmonization produced a unified matrix of 396 genera across all cohorts (retained at $\geq 20\%$ prevalence in at least one cohort, applied globally across all five cohorts before any supervised split), transformed to centered log-ratio (CLR) coordinates for compositional analysis; a dedicated sensitivity analysis re-deriving this feature set independently within each training split (never using held-out-cohort information) reproduced the same qualitative generalization-failure pattern (LOCO AUC changed by at most 0.066). The Zhu 2022 cohort deposited paired blood microbiome and fecal samples; only fecal samples ($n=90$) were retained for all analyses. Within four labeled cohorts ($n=401$ for binary AD/CN analyses), logistic regression and LightGBM classifiers achieved within-cohort AUC-ROC values ranging from 0.633 to 0.998 under nested cross-validation, broadly consistent with originally reported performance. For Zhu 2022, whose near-perfect within-cohort AUC (0.998 logistic regression, 0.979 LightGBM) is based on only 60 samples, 50 repetitions of the nested-CV partition gave a mean AUC of 0.9955 (SD 0.0027) for logistic regression and 0.9760 (SD 0.0103) for LightGBM, indicating this performance level is not an artifact of one particular fold assignment; a 1,000-permutation label-permutation test under the same strict training-only pipeline gave a null distribution centered near chance (mean AUC ≈ 0.47 for both models) with no permutation reaching the observed value for either model (empirical $p\approx 0.000999$, the resolution floor at this permutation count). In leave-one-cohort-out (LOCO) transfer experiments, LOCO AUC-ROC fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations collapsing to near chance (AUC = 0.504). A paired,

diagnosis-stratified, participant-level bootstrap (10,000 replicates, identical resampled participants used for both the within-cohort and LOCO score in each replicate) formally compared within-cohort and LOCO AUC for all eight cohort–model pairs: five of eight showed a delta reliably above zero (95% bootstrap CI on the delta excluding zero), while the remaining three (Zhuang 2018, both models; Kazakhstan LightGBM) showed the same-signed (within-cohort higher than LOCO) delta but with a 95% CI that included zero. PERMANOVA on Aitchison distance showed that cohort identity explained 19.3% of total Aitchison variance across all five cohorts ($F=30.13$, $p=0.0001$); in the four-cohort marginal model, cohort explained $R^2=0.172$ versus only 0.014 for disease label. This disparity is sensitive to cohort composition: restricting to the three technically more comparable Chinese paired-end MiSeq cohorts (Zhuang 2018, Ling 2020, Zhu 2022) reduced it from approximately 12-fold to approximately 2.6-fold (cohort $R^2=0.059$ vs. diagnosis $R^2=0.023$), and pairwise beta-dispersion comparisons showed the overall dispersion difference was driven predominantly by one cohort (Ling 2020) rather than uniformly across all pairs. Because cohort is a composite study-of-origin variable, PERMANOVA cannot decompose this variance into technical versus biological sources. Batch correction was evaluated under two designs: a label-blind design (ComBat-seq/MMUPHin fit without any diagnosis information) left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), while an exploratory label-informed design (fit using each sample’s own true diagnosis label, which cannot estimate prospective performance) produced large, method- and cohort-specific degradation, including a collapse to $AUC=0.308$ for one cohort under ComBat-seq — a result attributable in large part to the label-informed design itself rather than to batch correction per se. Neither design recovered cross-cohort transferability. SHAP analysis using out-of-fold values revealed near-zero pairwise Jaccard similarity between cohorts’ top-20 predictive taxa (mean 0.135 for logistic regression and 0.058 for LightGBM); against a null distribution matched to the observed statistic’s own construction (four random top-N sets per replicate drawn from the shared 396-genus pool, averaging the six pairwise Jaccards among them, 100,000 replicates), logistic regression overlap was significantly above chance at all three tested thresholds ($N=10$, 20, 50), while LightGBM was significant at $N=20$ and $N=50$ but not $N=10$. A further sensitivity analysis restricting each cohort’s random draws to genera that actually reach $\geq 20\%$ prevalence within that cohort’s own AD/CN samples (a per-cohort eligible universe rather than the shared pool) reversed the LightGBM finding: LightGBM overlap was not significant at any of $N=10$, 20, or 50 under this more realistic null ($p=0.80$, 0.24, 0.061), while logistic regression remained significant at all three thresholds. Using a pre-specified, descriptive cross-fold coefficient-sign-stability screen for logistic regression (positive or negative sign in ≥ 8 of 10 outer folds, not a formal hypothesis test), eight genera met the criterion for opposite, stable fitted-coefficient directions across cohorts; because LightGBM is a tree ensemble with no single coefficient-like direction, no comparable screen is applied to it. These results demonstrate substantial cross-cohort generalization limitations for gut microbiome-based AD classifiers among the evaluated publicly available 16S datasets, and that the batch correction methods evaluated here do not recover cross-cohort transferability. Multi-cohort harmonized training data and prospective validation across additional, more diverse cohorts are prerequisites before microbiome-based AD classifiers can be considered clinically actionable.

1. Introduction

Alzheimer's disease (AD) is the leading cause of dementia worldwide, affecting an estimated 55 million people, with projections exceeding 150 million by 2050 (World Health Organization, 2023). The gut–brain axis has emerged as a mechanistic pathway linking intestinal microbial communities to neurodegeneration: altered gut microbiome composition has been associated with AD pathology through neuroinflammatory, metabolic, and neuroendocrine routes (Cryan et al., 2019; Kowalski and Mulak, 2019). Since the landmark observation that germ-free APPPS1 transgenic mice showed reduced amyloid- β pathology in the absence of gut microbiota compared with conventionally colonized APPPS1 controls (Harach et al., 2017), dozens of 16S rRNA and shotgun metagenomic studies have reported significant differences in gut microbial composition between AD or MCI patients and cognitively normal (CN) controls. MCI is included as a diagnostic category in this literature given its role as a preclinical or prodromal stage of AD, though the primary supervised analyses in this study focus on binary AD versus CN comparisons (see Section 2.4).

The clinical appeal of a gut microbiome-based diagnostic signature is considerable: stool is non-invasive, inexpensive to collect, and microbiome profiling is increasingly accessible. Multiple independent research groups have trained machine learning classifiers on gut microbiome abundance profiles and reported promising within-cohort discrimination performance, with area under the receiver operating characteristic curve (AUC-ROC) values of 0.70–0.90 in several published studies (Zhuang et al., 2018; Ling et al., 2021; Zhu et al., 2022). However, a critical property of any diagnostic biomarker—transferability across independent populations—has rarely been tested in this literature.

The limited cross-cohort comparisons that do exist reveal a striking inconsistency in the direction of reported associations. A systematic review and meta-analysis synthesizing 17 published studies (679 AD/MCI patients, 632 controls) documented that *Bacteroides* abundance was significantly elevated in North American cohorts yet significantly reduced in Chinese cohorts—a directional reversal across geographically distinct populations (Jemimah et al., 2023). Similar contradictions have been observed for *Akkermansia*, *Faecalibacterium*, and *Lactobacillus* across individual studies (Vogt et al., 2017; Liu et al., 2019; Zhuang et al., 2018). These population-level contradictions suggest that published within-cohort classifiers may be detecting cohort-specific compositional signatures—driven by diet, geography, host genetics, sequencing protocols, or a combination thereof—rather than a universal microbial correlate of AD pathology.

A recent multi-cohort study integrating 15 public AD microbiome projects identified candidate gut microbiota indicators across multiple geographic populations using integrated data (Wang et al., 2026), confirming that multi-population integration remains a methodological frontier. The present study is complementary, providing a systematic quantification of cross-cohort generalization failure and an empirical evaluation of whether two commonly used batch-correction approaches recover cross-cohort transferability—questions that remain open even where multi-cohort classifiers have been attempted.

Despite this recognized heterogeneity, few studies have jointly quantified: (1) the magnitude of cross-cohort AUC collapse when trained classifiers are applied to independent cohorts; (2) the relative magnitude of cohort-associated versus diagnosis-associated compositional variance, as estimated by variance partitioning — noting that “cohort” is itself a composite, study-of-origin variable (bundling sequencing platform, library preparation, recruitment site, diagnostic ascertainment, geography, and diet, among other differences) that a variance-partitioning method such as PERMANOVA cannot decompose into technical and biological components; and (3) whether state-of-the-art microbiome batch correction methods can recover cross-cohort generalization. Answering these three questions jointly in one rigorous analysis is the primary contribution of this paper.

We assembled five publicly available 16S rRNA gene sequencing cohorts spanning three countries (China, Kazakhstan, South Korea) and conducted a comprehensive cross-cohort generalization evaluation. We report that: classifiers showed substantial and variable degradation when transferred to independent cohorts; cohort identity explains substantially more compositional variance than disease status, though the size of this disparity depends materially on which cohorts are compared; under label-blind transductive adjustment, ComBat-seq and MMUPHin left mean cross-cohort AUC essentially unchanged and did not recover transferability; and top predictive taxa showed low absolute overlap across cohorts, with a subset of taxa showing opposite, stable fitted-coefficient directions across cohorts. Together, these findings constitute a systematic quantification of cross-cohort generalization limitations among the evaluated publicly available gut microbiome AD classifier datasets, with direct implications for the field's trajectory toward clinical translation.

2. Materials and Methods

2.1 Cohort Identification and Data Acquisition

We performed a systematic search of NCBI SRA and ENA for publicly available 16S rRNA amplicon sequencing studies of gut microbiota in AD, MCI, or cognitively normal populations, deposited before January 2025. The search used the following terms in the SRA search interface: (“Alzheimer’s disease” OR “Alzheimer disease” OR “mild cognitive impairment” OR “MCI” OR “dementia”) AND (“gut microbiome” OR “gut microbiota” OR “intestinal microbiome” OR “16S rRNA”) AND (“Homo sapiens”). Studies were included if they provided: (1) per-sample diagnosis labels (AD, MCI, or CN) accessible in public metadata; (2) raw FASTQ files or quality-filtered reads deposited in a public archive; (3) minimum sample sizes of 40 participants per cohort. We restricted this analysis to 16S rRNA studies; shotgun metagenomic cohorts were identified but excluded to avoid the additional platform confound introduced by mixing amplicon and whole-genome sequencing data. Cohorts were excluded if: per-sample labels were IRB-restricted (for supervised analyses); data had not been publicly released at the time of analysis; or samples represented a tissue compartment other than fecal microbiome as the primary analyte. A complete record of the search procedure, identified cohorts, and exclusion rationale is provided in Supplementary Note S1.

Four cohorts met all criteria for supervised analysis. A fifth cohort, Kim/KBASE 2022 (PRJEB50447), lacked publicly available per-sample diagnosis labels and was retained only for unsupervised cohort-level analyses. The ALBION 2025 Greek cohort (PRJNA1297934) was identified but excluded because data had not yet been publicly released at the time of analysis. Group-level counts from the published paper (18 amyloid-positive preclinical AD; 60 cognitively normal) were available but insufficient for supervised classifier training or evaluation.

The four cohorts with per-sample diagnosis labels used in supervised analyses were:

Zhuang 2018 (China; PRJNA554111): 86 participants (43 AD, 43 CN) from the Third Military Medical University, Chongqing. Paired-end MiSeq 2×300 bp, V3–V4 region (Zhuang et al., 2018).

Ling 2020 (China; PRJNA633959): 171 participants (100 AD, 71 CN). Paired-end MiSeq 2×300 bp, V3–V4 region (Ling et al., 2021).

Zhu 2022 (China; PRJNA489760): 180 samples deposited in SRA from a larger 302-participant study, representing 90 participants (30 AD, 30 MCI, 30 CN) each contributing 2 sample types = 180 samples. Each participant contributed two samples: a fecal (F_) *sample and a blood microbiome (B_)* sample. Only the 90 fecal samples (30 AD, 30 MCI, 30 CN) were used in all analyses; the 90 blood microbiome samples were excluded as they represent a distinct biological compartment and would confound within-cohort classifiers. Binary analyses used 30 AD and 30 CN fecal samples (n=60). Paired-end MiSeq 2×300 bp, V3–V4 (Zhu et al., 2022). Diagnostic criteria across cohorts relied on clinical AD diagnosis (Zhuang, Ling, Zhu) or amyloid-PET positivity (Kim); these differences in diagnostic ascertainment are acknowledged as a source of between-cohort biological heterogeneity.

Kazakhstan 2022 (PRJNA811324): 84 participants (41 AD, 43 CN). Single-end NovaSeq sequencing, V3–V4 region (Kaiyrlykzy et al., 2022). This cohort is the only single-end dataset in the study; implications for read length and taxonomic resolution are discussed in Section 2.2 and the Limitations section.

The Kim/KBASE 2022 cohort (Jung et al., 2022; PRJEB50447; 78 participants; 18 preclinical AD by amyloid-PET, 60 CN) was included in PERMANOVA analyses (Phase 4) only and is explicitly excluded from all supervised classifier training and evaluation.

All raw FASTQ data were downloaded from NCBI SRA using fasterq-dump (SRA Toolkit v3.0.10). Cohort-level sequencing statistics are reported in Supplementary Table S2.

2.2 DADA2 Amplicon Sequence Variant Processing

Raw paired-end reads (four cohorts) and single-end reads (Kazakhstan) were processed independently per cohort using DADA2 v1.28 in R v4.3 (Callahan et al., 2016), to avoid introducing cross-cohort data leakage during error model estimation.

For paired-end cohorts, reads were trimmed to truncation lengths selected per cohort based on per-base quality score profiles (generally 240 bp forward / 200 bp reverse for V3–V4 MiSeq 2×300 data), filtered to remove reads with >2 expected errors, chimeras were removed using the consensus method, and forward and reverse reads were merged. For Kazakhstan (single-end), the single-end DADA2 workflow was applied without merging; truncation length was adjusted to 250 bp after inspection of quality profiles (see Supplementary Note S2 for rationale). The mixed single-end/paired-end design is an additional technical source of between-cohort heterogeneity acknowledged as a study limitation.

Taxonomic assignment used the SILVA reference database v138.1. ASVs were collapsed to genus level by summing raw counts per sample. A genus was retained across the entire study if it reached $\geq 20\%$ prevalence (i.e., a nonzero count in $\geq 20\%$ of samples) in at least one of the five cohorts; this prevalence threshold was computed per cohort and the union of genera passing in any cohort was taken across all five cohorts, producing a unified feature set of 396 genera. Genera absent in a given sample received a count of zero. Because this feature-selection step used all five cohorts' raw prevalence simultaneously, the resulting 396-genus feature universe was informed, in part, by the prevalence of the eventual test cohort in every downstream leave-one-cohort-out (LOCO) and pairwise-transfer experiment (Section 2.5). This is not a leakage-free design, and we do not describe it as one; it is transparently identified as a global, all-cohort feature-selection step, and its effect on the paper's conclusions is evaluated directly by a strict training-only sensitivity analysis (Section 2.5.1) rather than assumed to be negligible.

Leakage status of every preprocessing step, stated explicitly. To avoid any ambiguity about which steps could, in principle, leak held-out-cohort or held-out-sample information into a downstream evaluation, every preprocessing operation used in this paper is classified here: (1) *DADA2 ASV inference and error-model estimation* (this section) is performed independently per cohort using only that cohort's own reads, is label-free (uses no diagnosis information), and cannot leak across cohorts or across AD/CN status by construction. (2) *Global genus-level feature selection* ($\geq 20\%$ prevalence, this section) pools raw prevalence across all five cohorts, including the eventual held-out cohort in every LOCO/pairwise experiment; this is a genuine, acknowledged methodological limitation, not leakage-free, and its practical impact is bounded by the training-only sensitivity analysis in Section 2.5.1, which re-derives the feature set from only the training-side samples of every individual split and finds the qualitative generalization-failure conclusion unchanged. (3) *Zero-replacement and CLR transformation* (Section 2.3) are deterministic, closed-form functions applied independently to each sample's own count vector; they use no information from any other sample, cohort, or diagnosis label, and are therefore sample-local operations with no leakage pathway of any kind. (4) *Model hyperparameter tuning and fitting* (Sections 2.4–2.5) use only the training-side samples of whichever split is being evaluated (inner CV within a training fold for within-cohort analyses; the training-cohort pool for LOCO/pairwise analyses), with test-side samples never used for tuning, fitting, or feature standardization. (5) The *strict training-only sensitivity re-derivation* (Section 2.5.1) additionally re-derives genus selection itself from only the training-side samples of every split, closing the one leakage pathway identified in (2) for that specific analysis.

2.3 Compositional Data Transformation

Raw genus-level count tables were converted to centered log-ratio (CLR) coordinates for all downstream analyses requiring distributional assumptions on unconstrained data (machine learning classifiers, PERMANOVA on Euclidean distance). The CLR transformation is defined as:

$$\text{clr}(x_i) = \log \left(\frac{x_i}{g(x)} \right)$$

where $g(x)$ is the geometric mean of the composition x .

Zero counts were replaced prior to transformation using the multiplicative replacement strategy (Martín-Fernández et al., 2003) with $\delta = 0.65$:

$$x_i^* = \begin{cases} \delta & \text{if } x_i = 0 \\ (1 - n_0 \cdot \delta / T) \cdot x_i & \text{if } x_i > 0 \end{cases}$$

where $T = \sum_j x_j$ is the total read count of the sample and n_0 is the count of zeros in the sample. The unified genus matrix contained 75.4% zero entries (151,926 of 201,564 cells across the 509 fecal samples and 396 genera). The value $\delta = 0.65$ was chosen such that the replacement mass remained small relative to typical per-sample sequencing depth (median 29,521 reads): in the worst-observed sample, the total mass allocated to zero-filled entries was 2.53% of that sample's total reads. A built-in safeguard rescales δ downward whenever $n_0 \cdot \delta \geq T$, capping replacement mass at 1% of sample total in such cases. This strategy preserves the ratios among non-zero components while introducing a minimal perturbation proportional to the relative frequency of zeros, and is preferable to the commonly used additive pseudocount which distorts low-abundance taxa more severely.

CLR-transformed values were used as features for all machine learning models and for Aitchison distance computation (Euclidean distance on CLR values). Raw genus-level count matrices (not CLR-transformed) were provided as input to ComBat-seq, which requires non-negative integer counts. Relative abundances (proportions) were provided to MMUPHin. Bray-Curtis dissimilarity was computed on relative abundances. CLR values were used for all other analyses.

2.4 Within-Cohort Classifier Baselines

We trained binary classifiers (AD vs. CN) independently for each of the four labeled cohorts using two model families:

L2-regularized logistic regression (scikit-learn v1.5; Pedregosa et al., 2011; LogisticRegression, solver="saga"). Regularization strength C was tuned over {0.001, 0.01, 0.1, 1.0, 10.0} in inner cross-validation. An `l1_ratio` parameter is present in the estimator configuration but has no effect under the L2 penalty actually used (scikit-learn's `penalty` argument was not set to "elasticnet"); all logistic-regression results in this paper are L2 (ridge)-regularized, not elastic-net.

LightGBM gradient-boosted trees (lightgbm v4; Ke et al., 2017). Hyperparameters tuned: number of estimators $\in \{50, 100, 200\}$, learning rate $\in \{0.05, 0.1\}$, max depth $\in \{3, 5\}$, number of leaves $\in \{15, 31\}$.

For Zhu 2022, AD was defined as the positive class and MCI samples were excluded from binary analyses (resulting in n=60: 30 AD, 30 CN fecal samples); a separate three-class analysis (AD / MCI / CN) used all 90 fecal samples and is reported as a secondary finding. Because only fecal samples were retained, each patient appears exactly once; standard StratifiedKFold was used without any paired-sample grouping structure.

Performance was estimated using nested cross-validation: a 10-fold outer loop for performance estimation and a 5-fold inner loop for hyperparameter tuning. The primary metric was AUC-ROC computed on out-of-fold predictions pooled across the 10 outer folds, providing a single unbiased AUC estimate per cohort-model combination. Bootstrap 95% confidence intervals (1,000 resamples) on the pooled out-of-fold predictions are reported throughout. We also report sensitivity and specificity at the Youden's J optimal threshold. For comparison with originally reported performance, the published AUC values from the original cohort papers are noted in Table 2 where available.

2.4.1 Zhu 2022 Robustness: Repeated Cross-Validation and Label Permutation

Because Zhu 2022's near-perfect within-cohort AUC (Section 3.2) is estimated from only 60 samples (30 AD, 30 CN), we assessed its robustness with two additional analyses, both run under the strict training-only feature-selection pipeline (Section 2.5.1) rather than the global feature set, so that no step of either analysis could leak held-out-fold information.

Repeated nested cross-validation. The 10-fold outer / 5-fold inner nested-CV procedure (Section 2.4) was repeated 50 times, each repetition using a different, pre-specified deterministic random seed (seeds 1000–1049) for the stratified outer-fold partition, with genus selection, zero-replacement, CLR transformation, and model tuning re-derived independently within every outer fold of every repetition using only that fold's training-side samples. We report the mean, standard deviation, median, 2.5th/97.5th percentile, minimum, and maximum pooled-OOF AUC across the 50 repetitions. This procedure quantifies sensitivity to the arbitrary choice of fold assignment; it is explicitly not a set of independent replications of

the study (the underlying 60 participants are identical across all 50 repetitions), and we report it as fold-assignment stability rather than as an independent-sample validation.

Label permutation test. Under the same strict training-only pipeline and the same fixed outer-fold partition (random_state=42, matching the partition used throughout the rest of this paper), we additionally ran a label-permutation test: in each of exactly 1,000 permutations, the AD/CN labels were randomly shuffled across the 60 participants (preserving the 30/30 class split) using a pre-specified seed, and the entire training-only pipeline — genus selection, zero-replacement, CLR transformation, hyperparameter tuning, and model fitting — was re-run inside every outer fold using only that fold’s training-side samples and the permuted labels, producing a null pooled-OOF AUC for that permutation. Because genus selection and CLR feature construction depend only on which participants are in a fold’s training set (not on their labels), these steps were precomputed once per outer-fold partition and reused across all permutations — a pure implementation optimization with no effect on the statistical result, since the model itself is always refit on the permuted labels. The empirical p-value was computed as $(1 + \# \{\text{null AUC} \geq \text{observed AUC}\}) / (1000 + 1)$, using the training-only single-partition AUC as the observed value; the minimum attainable p-value at this permutation count is $1 / 1001 \approx 0.000999$, and we do not interpret any p-value below this floor as more precise than the floor itself.

2.5 Cross-Cohort Generalization Experiment

We evaluated two cross-cohort transfer designs:

Leave-one-cohort-out (LOCO): For each test cohort, a classifier was trained on the pooled samples from the remaining three labeled cohorts (no cross-validation within this training step; hyperparameters were selected via 5-fold inner CV on the training pool). The trained model was then evaluated on the held-out cohort. This produced four LOCO AUC estimates per model type. We report the LOCO AUC drop as the difference between within-cohort (nested CV) AUC and LOCO AUC; bootstrap 95% CIs are reported for each LOCO AUC estimate. Non-overlap of two independently computed confidence intervals is not a formal statistical test of whether their difference is nonzero, and is not used as one anywhere in this paper; the AUC drop is instead evaluated with the paired bootstrap procedure described in Section 2.5.2.

Pairwise transfer: A classifier was trained on one cohort and tested on each of the other three cohorts individually, yielding a 4×4 matrix (diagonal excluded) of cross-cohort AUC values per model type. This design is more stringent than LOCO and isolates specific directional failures.

Bootstrap 95% CIs (1,000 resamples) on the test set predictions are reported for all LOCO and pairwise estimates.

Sensitivity analyses: To assess whether generalization failure was attributable to any single cohort, we repeated the LOCO experiment under three additional configurations: (A) excluding Kazakhstan entirely, with LOCO evaluated on the remaining three cohorts; (B) excluding Zhuang 2018; (C) excluding Ling 2020. Results are reported in Section 3.3 and Supplementary Figure S4.

2.5.1 Training-Only Feature-Selection Sensitivity Analysis

Because the 396-genus feature set (Section 2.2) was selected globally using all five cohorts’ raw prevalence, every LOCO and pairwise experiment above used a feature universe partly informed by the eventual held-out cohort. To assess whether this affected the generalization-failure conclusion, we repeated

the within-cohort nested-CV, LOCO, and pairwise-transfer analyses with genus selection re-derived independently inside every individual split, using only that split’s training-side samples: for each outer fold (within-cohort), each held-out cohort (LOCO), and each train cohort (pairwise), the $\geq 20\%$ -prevalence filter was recomputed from scratch using only the training-side AD/CN-labeled samples for that specific split, and the corresponding test/held-out samples were projected onto the training-derived genus set (absent genera filled with zero count) before per-sample zero-replacement and CLR transformation. Model families, hyperparameter grids, random state, fold counts, and the bootstrap CI procedure were identical to the main analyses (Sections 2.4–2.5). This produces a strictly train-only feature universe for every split, at the cost of a smaller and split-specific genus count (e.g., 150 genera when Kazakhstan is excluded from feature selection during its own LOCO hold-out, versus 396 in the main global-feature analysis).

2.5.2 Formal Within-Cohort vs. LOCO AUC Difference

Non-overlapping bootstrap confidence intervals are a heuristic, not a formal statistical comparison of two AUC values. To formally evaluate the within-cohort-versus-LOCO AUC difference for each of the eight cohort–model combinations, we performed a paired, diagnosis-stratified, participant-level bootstrap. For each cohort–model pair, the within-cohort out-of-fold prediction and the LOCO (that cohort held out) prediction refer to the same participants; each of 10,000 bootstrap replicates resampled participants with replacement, separately within the AD group and within the CN group, once per replicate, and used that same resampled index set to compute both the within-cohort AUC and the LOCO AUC — so the resulting distribution of $\Delta = \text{AUC}_{\text{within}} - \text{AUC}_{\text{LOCO}}$ reflects a genuine paired comparison rather than the difference of two independently resampled statistics. We report the observed delta, the bootstrap mean delta, the 95% percentile CI of the bootstrap delta distribution, and the proportion of bootstrap replicates with $\Delta \leq 0$; this proportion is an empirical bootstrap quantity, not an exact parametric p-value, and we do not describe it as one. Every within-cohort and LOCO AUC 95% CI reported elsewhere in this paper (Table 2, Table 3, Table S1) was regenerated using this same diagnosis-stratified, participant-level bootstrap procedure (single-statistic percentile CI, 10,000 replicates), so that every confidence interval in the paper is produced by one consistent, explicitly specified resampling procedure.

Within-cohort out-of-fold prediction files do not carry an explicit participant identifier; participant identity was recovered by replicating the exact data-loading and filtering sequence used to produce them and verified, before use, by an exact-match assertion between the recovered diagnosis vector and the file’s own saved outcome column, for every cohort–model combination.

2.6 PERMANOVA Variance Decomposition

To quantify the relative contribution of cohort identity and disease status to total compositional variance, we conducted PERMANOVA (permutational multivariate analysis of variance) using the `adonis2` function in the R package `vegan` v2.6 (Oksanen et al., 2024). Two distance matrices were used:

- **Aitchison distance** (Euclidean distance on CLR values): the compositionally appropriate distance that does not distort due to the simplex constraint.
- **Bray-Curtis dissimilarity** (on relative abundances): included for comparability with the broader microbiome literature.

All PERMANOVA models used 9,999 permutations with Type III (marginal) sums of squares (by="margin") to allow independent assessment of cohort and diagnosis effects. The minimum attainable permutation p-value with 9,999 permutations is reported as $p=0.0001$ throughout. Three models were fitted: (1) cohort only, all five cohorts ($n=509$); (2) cohort + diagnosis, restricted to the four labeled cohorts ($n=401$: AD and CN samples only, Zhu 2022 MCI excluded), with marginal R^2 for each term; (3) within each labeled cohort, diagnosis as the sole predictor (within-cohort disease effect size). Note that Figure 4B displays the PCoA from Model 1 ($n=509$, all five cohorts including MCI samples), while the marginal R^2 values reported throughout the text derive from Model 2 ($n=401$).

Homogeneity of multivariate dispersions was tested using betadisper and permutest (9,999 permutations) to assess whether observed cohort differences reflected centroid separation, within-group spread heterogeneity, or both. A significant beta-dispersion test indicates that PERMANOVA R^2 values partially reflect variation in within-group spread, not only centroid shifts. To identify which specific cohort pairs drive the overall dispersion difference, we additionally computed all pairwise betadisper comparisons among the five cohorts (Aitchison distance, 9,999 permutations per pair), with Holm and Benjamini-Hochberg multiplicity adjustment applied across the 10 pairwise p-values.

Cohort is a composite variable; PERMANOVA cannot decompose it into technical and biological sources. "Cohort" in this study bundles sequencing platform, library preparation, recruitment site and country, clinical diagnostic ascertainment, and unmeasured dietary and lifestyle differences into a single categorical variable. A PERMANOVA or betadisper term for cohort therefore estimates the total compositional variance associated with study-of-origin, not a technical-batch-effect estimate specifically; no method used in this paper decomposes that total into its technical and biological components. To assess how sensitive the cohort-versus-diagnosis variance disparity (Section 3.4) is to this composite structure, we repeated the marginal PERMANOVA and betadisper analyses on a subset restricted to the three cohorts that are more technically comparable to one another — Zhuang 2018, Ling 2020, and Zhu 2022, all Chinese, paired-end MiSeq 2×300bp, V3–V4 studies — while still differing in recruitment site, exact diagnostic procedure, and other unmeasured factors, so this subset is more technically comparable, not technically identical. This sensitivity analysis used the same marginal (by="margin") two-variable (cohort + diagnosis) PERMANOVA specification and 9,999 permutations as the main four-cohort model, restricted to binary AD/CN samples from the three cohorts ($n=317$), together with betadisper/permutest and pairwise post-hoc comparisons on the same subset.

To further quantify the degree to which cohort-specific information is retained in the preprocessed feature space, we trained a logistic regression classifier (one-vs-rest, stratified 5-fold CV) to predict cohort identity from the CLR-transformed features across all 509 samples. The macro-AUC for cohort prediction provides a complementary measure of cohort separability.

2.7 Batch Correction

We evaluated one count-based batch-correction method (ComBat-seq) and one microbiome-specific meta-analysis framework (MMUPHin), applying each with cohort identity as the batch variable, under two distinct designs that differ in whether diagnosis (disease-label) information is supplied during correction. Both designs are **transductive**: both methods estimate batch-effect parameters jointly across every sample being corrected — including, in the downstream LOCO evaluation, the eventual held-out test cohort's own feature values and cohort membership. Neither design is prospective or inductive (i.e., neither defines a

transformation that could be applied to a genuinely new, unseen patient without refitting); they differ only in whether the target cohort’s diagnosis labels are additionally used.

Design 1 — label-blind transductive correction (primary). ComBat-seq (Zhang et al., 2020) was run on raw genus-level count matrices (genera $\times$ samples, four labeled cohorts only) with `group = NULL`, `full_mod = FALSE` — no diagnosis information of any kind is supplied. MMUPHin v1.15 (Ma et al., 2022) was run on relative-abundance profiles from the four labeled cohorts with `batch = cohort` and `covariates = NULL`; the metadata object supplied to `adjust_batch` did not include a diagnosis column at all. In both cases, cohort identity and every sample’s own raw feature values are used (this is what makes the design transductive), but no sample’s true AD/CN label is ever used during correction. Corrected count tables (ComBat-seq) or relative abundances (MMUPHin) were CLR-transformed using the same multiplicative zero-replacement procedure described in Section 2.3.

Design 2 — label-informed transductive sensitivity analysis (exploratory). The same two methods were additionally run with `group = disease_status`, `full_mod = TRUE` (ComBat-seq) and `covariates = diagnosis` (MMUPHin) — i.e., using every sample’s own true diagnosis label, for all four labeled cohorts jointly, including the sample’s own held-out-cohort membership in the eventual LOCO evaluation. This design cannot estimate prospective external-validation performance, because the correction step is fit using every sample’s own diagnosis label, including test-cohort samples. We retain it only as an exploratory sensitivity analysis illustrating an upper/alternative bound on what correction with label information could do, not as evidence about deployable performance.

After each correction, the LOCO generalization experiment (Section 2.5) was repeated on the corresponding batch-corrected CLR matrices, and within-cohort AUC on corrected data was re-computed under the same nested-CV procedure as Section 2.4. Because Design 2’s correction step is explicitly told each sample’s own diagnosis label as part of a global, jointly-fit model, a within-cohort AUC measured on Design-2-corrected features cannot be used to argue that “genuine” disease-associated biological signal survived correction independent of that label information — some or all of any preserved within-cohort separability could instead be an artifact of the correction algorithm’s own label-preservation mechanism (`full_mod=TRUE / covariates="diagnosis"`) rather than evidence about biology. This caveat does not apply to Design 1, whose within-cohort AUC on corrected data is not confounded by label use during correction, though it remains a transductive (not prospective) result like the LOCO evaluation itself.

2.8 SHAP Feature Importance, Direction, and Cross-Cohort Overlap Analysis

To identify which genera drove predictions in each cohort and whether these features were consistent across cohorts, we computed SHapley Additive exPlanations (SHAP) values (Lundberg and Lee, 2017) using an out-of-fold (OOF) procedure to avoid the in-sample overfitting bias that arises when SHAP values are computed on the same data used to fit the model.

For each labeled cohort, SHAP values were computed using the same 10-fold outer CV structure used for AUC estimation (Section 2.4). In each fold, the classifier was trained on the 9 training folds; SHAP values were then computed for the held-out fold using the training fold as the SHAP background distribution. For logistic regression, `shap.LinearExplainer` was used with the training-fold data as background (`feature_perturbation="interventional"`), and the held-out fold as the evaluation set; this ensures the mean SHAP value over the evaluation set is not algebraically forced to zero (as would occur if background and evaluation sets were identical). For LightGBM, `shap.TreeExplainer` was used (`check_additivity=False`)

with the same OOF structure. SHAP values were aggregated across all 10 folds: the mean absolute SHAP value per genus ($|\text{SHAP}|$) was used as the importance metric for both models, and is not affected by the direction issue described below.

Direction is not defined from mean signed SHAP. For a linear model evaluated with `LinearExplainer` under interventional feature perturbation, the SHAP value for feature j is approximately $\beta_j \cdot (x_{ij} - \bar{x}_j^{\text{background}})$, so the mean signed SHAP value over an evaluation set is approximately $\beta_j \cdot (\bar{x}_j^{\text{eval}} - \bar{x}_j^{\text{background}})$ — a quantity that depends on both the fitted coefficient *and* the mean-abundance difference between the evaluation set and whichever background set was chosen, not on the coefficient sign alone. Within a single cohort’s own OOF folds this background-vs-eval difference is close to sampling noise (both come from the same cohort), but it is not zero, and for cross-cohort (LOCO) SHAP the background (the $N-1$ training cohorts) and evaluation set (the held-out cohort) come from systematically different populations by construction — exactly the compositional difference quantified by PERMANOVA in Section 2.6 — so mean signed LOCO SHAP is confounded with raw cross-cohort abundance differences rather than reflecting the fitted model’s direction. For this reason, mean signed SHAP is not used anywhere in this paper as a global, cross-cohort direction or biological-association statistic.

Coefficient-based direction and stability (logistic regression only). For logistic regression, direction is instead defined from the fitted model’s own coefficient. In each of the 10 outer OOF folds per cohort, the coefficient for every genus was recorded, together with its sign. A genus’s direction in a given cohort was called **stable** — positive or negative — if the coefficient carried that sign in at least 8 of the 10 outer folds (a threshold specified before any results were examined); a coefficient of exactly zero counted toward neither direction. A genus was classified as meeting this **descriptive coefficient-sign-stability screen** only if it (a) appeared in the top-20 OOF mean- $|\text{SHAP}|$ ranking in at least two cohorts, (b) had a stable positive coefficient in at least one of those cohorts, and (c) had a stable negative coefficient in at least one other. This is a conservative definition relative to a naive sign check on mean SHAP, because it requires the direction to be reproducible across most of the 10 outer folds within each cohort, not merely to differ in its single pooled mean-SHAP sign. **We treat this screen as descriptive, not as a formal hypothesis test: the 10 outer-CV folds within a cohort share substantially overlapping training data (each fold trains on the same 9/10 of that cohort’s samples), so fold-level coefficient signs are not independent statistical replications, and no multiplicity-adjusted p-value is assigned to any individual taxon. LightGBM is a tree ensemble with no single coefficient-like, globally monotonic direction, and its SHAP feature-genus relationships may be nonlinear or non-monotonic; no coefficient-based (or any other) global direction or stability screen is therefore applied to LightGBM anywhere in this paper.** LightGBM results are limited to OOF mean- $|\text{SHAP}|$ feature importance and the top-feature-overlap/Jaccard analysis below, which do not require a direction assignment.

For LOCO analyses, SHAP values were computed on the held-out test cohort from each LOCO-trained model, with the three-cohort training pool as the SHAP background, and are reported and used only as a feature-*importance* ranking (mean $|\text{SHAP}|$); consistent with the confound described above, no cross-cohort AD/CN direction is assigned to LOCO SHAP values for either model.

Cross-cohort taxonomic overlap. Pairwise taxonomic overlap between cohorts was quantified using Jaccard similarity on the top- N SHAP genera ($N \in \{10, 20, 50\}$); $N=20$ is the primary threshold. The statistic actually reported is the mean of the six unique pairwise Jaccard values among the four labeled

cohorts' top-N sets ($\binom{4}{2} = 6$ pairs). Two null distributions are reported. The primary, matched-statistic null was constructed to match the observed statistic's own construction exactly: in each of 100,000 replicates, four independent random top-N gene sets were drawn (without replacement) from the full, shared 396-genus pool, and the same six unique pairwise Jaccard similarities among those four random sets were averaged to produce one null replicate. The empirical one-sided p-value was computed as $(1 + \# \{ \text{null} \geq \text{observed} \}) / (n_{\text{perm}} + 1)$ (Davison and Hinkley, 1997).

A second, prevalence-restricted null addresses a limitation of drawing from the shared 396-genus pool: per-cohort genus prevalence varies substantially (e.g., Kazakhstan alone reaches $\geq 20\%$ prevalence for 356 of 396 genera, versus 115–131 for the three MiSeq cohorts), so a genus essentially absent from a given cohort could still be drawn into that cohort's random comparison set under the shared-pool null. For this sensitivity analysis, each cohort's own top-N ranking and its random draws were both restricted to genera reaching $\geq 20\%$ prevalence within that specific cohort's own AD/CN-labeled samples — a per-cohort eligible universe rather than the shared pool — with 100,000 replicates per (model, N) combination and every cohort confirmed to have at least N eligible genera before running (no adaptation of N was required for $N \in \{10, 20, 50\}$). We report both nulls and do not assume that significance under the shared-pool null implies significance under the prevalence-restricted null, or vice versa. A genus was classified as a directional flip using the coefficient-stability criterion above, not from this overlap analysis.

2.9 AI Tool Disclosure

Initial manuscript drafts and computational scripts were prepared with minimal assistance from Claude Sonnet 5 (Anthropic). All analytical decisions, parameter choices, experimental design, statistical interpretations, and scientific conclusions were made by the author. All code and text were reviewed, validated, and substantially revised under the author's direction.

2.10 Statistical Analysis Environment

DADA2 preprocessing (Section 2.2) was run using DADA2 v1.28 in R v4.3 and was not rerun for any analysis reported in this paper. The initial classifier, PERMANOVA, batch-correction, and SHAP analyses (Sections 2.4–2.8, excluding the additions described below) used scikit-learn v1.5, lightgbm v4, shap v0.51, pandas v2.x, and numpy v1.x in a conda environment (Python 3.11), and vegan v2.6, sva v3.50 (ComBat-seq), and MMUPHin v1.15 in R v4.3. Analyses added in response to peer review — the training-only feature-selection sensitivity, label-blind batch correction, SHAP coefficient-stability screen, the matched-statistic Jaccard null (Sections 2.5.1, 2.7, 2.8), the Zhu 2022 repeated-CV and permutation robustness analyses (Section 2.4.1), the paired within-cohort-vs-LOCO AUC-difference bootstrap (Section 2.5.2), the pairwise beta-dispersion and Chinese-MiSeq PERMANOVA sensitivity analyses (Section 2.6), and the prevalence-restricted Jaccard null (Section 2.8) — were executed in Python 3.11.15 (scikit-learn 1.5.2, lightgbm 4.6.0, shap 0.51.0, matching the original Python package versions) and R 4.5.1 (vegan 2.7.1, sva 3.58.0, MMUPHin 1.24.0, newer than the initial R environment, reflecting routine system updates over the course of this project). This mixed provenance — original DADA2/R4.3 preprocessing feeding a downstream analysis that combines an unchanged Python environment with an updated R4.5.1 environment — is disclosed here rather than described as a single uniform environment. Figures were generated with matplotlib v3.8 or v3.11 at 300 DPI.

3. Results

3.1 Cohort Characteristics and Processing

Five cohorts comprising 599 deposited samples from three countries were assembled (Figure 1A; Table 1). After DADA2 processing and chimera removal, read retention rates varied substantially across cohorts (47.9%–78.3%; Supplementary Table S2), reflecting differences in sequencing platform, library preparation, and amplicon length. The Kazakhstan cohort (single-end, NovaSeq) had notably different quality dynamics compared to the paired-end MiSeq cohorts. SILVA v138.1 genus-level classification and cross-cohort union produced a unified feature matrix of 396 genera. The 90 blood microbiome (B_*) samples from Zhu 2022 were excluded prior to all analyses (Section 2.4), leaving a working dataset of 509 samples across five cohorts. After construction of the global 396-genus feature schema, multiplicative zero replacement and CLR transformation were applied independently to each of the 509 samples, at $\delta=0.65$.

For supervised classifier analyses (Phases 2–3 and 5–6), the Kim/KBASE 2022 cohort was excluded because per-sample diagnosis labels are IRB-restricted (Supplementary Note S1), leaving four labeled cohorts with a total of 401 samples for binary AD/CN analyses (86 Zhuang + 171 Ling + 60 Zhu fecal binary subset + 84 Kazakhstan; Zhu 2022 MCI samples excluded from binary analyses).

3.2 Within-Cohort Classification Performance

Within-cohort nested cross-validation yielded AUC-ROC values broadly consistent with those reported in the original publications (Figure 2; Table 2). Logistic regression achieved AUC-ROC of 0.633 [95% CI: 0.51–0.75] for Zhuang 2018, 0.865 [0.81–0.92] for Ling 2020, 0.998 [0.99–1.00] for Zhu 2022, and 0.766 [0.66–0.86] for Kazakhstan. LightGBM yielded similar values: 0.635 [0.52–0.75], 0.862 [0.80–0.91], 0.979 [0.94–1.00], and 0.694 [0.57–0.80] respectively.

The near-perfect within-cohort AUC for Zhu 2022 fecal samples (logistic regression 0.998; LightGBM 0.979; $n=60$ binary) is notable given the small sample size (30 per class). With 10-fold CV on 60 samples, each test fold contains approximately six samples; fold-level AUC estimates can be unstable at this scale. PERMANOVA analysis (Section 3.4) reveals that diagnosis explains $R^2=0.094$ of within-cohort compositional variance ($F=6.01$, $p=0.0001$), suggesting statistically detectable diagnosis-associated compositional differences in this cohort. However, the high AUC should be interpreted cautiously and the LOCO experiment (Section 3.3) provides a more informative assessment of how transferable this signal is.

Robustness to fold assignment and to label permutation (Section 2.4.1). Repeating the strict training-only nested-CV partition 50 times under different pre-specified seeds gave a mean pooled-OOF AUC of 0.9955 (SD 0.0027, range 0.9889–0.9989) for logistic regression and 0.9760 (SD 0.0103, range 0.9378–0.9944) for LightGBM (Supplementary Table S8; Supplementary Figure S6) — closely matching the single-partition training-only values (0.9956 and 0.9700, respectively) and showing that the near-perfect AUC is not an artifact of one particular fold assignment. This is a fold-assignment stability check, not an independent replication of the underlying 60-participant sample, which is identical across all 50 repetitions. A complementary label-permutation test under the same strict training-only pipeline and fixed outer-fold partition (1,000 permutations; Section 2.4.1) produced a null AUC distribution centered near chance for both models (logistic regression: null mean 0.473, SD 0.102, 95% range [0.280, 0.673]; LightGBM: null mean 0.472, SD 0.109, 95% range [0.264, 0.688]) — well below the observed training-only single-partition values of 0.9956 and 0.9700 respectively (Supplementary Table S9; Supplementary

Figure S6). No permutation among the 1,000 drawn reached the observed AUC for either model, giving an empirical p-value at the resolution floor for this permutation count ($p = 1/1,001 \approx 0.000999$ for both models); we report this p-value at its actual resolution and do not interpret it as more precise than this floor.

The three-class analysis (AD / MCI / CN) on all 90 Zhu 2022 fecal samples using logistic regression (one-vs-rest macro-AUC under nested cross-validation) yielded macro-AUC=0.761, confirming the multi-class discriminability of this cohort’s fecal compositional data.

For Zhuang 2018, within-cohort AUC values (0.633–0.635) were modest, consistent with the small sample size and weak within-cohort diagnosis effect observed in PERMANOVA ($R^2=0.017$, $p=0.066$; Section 3.4).

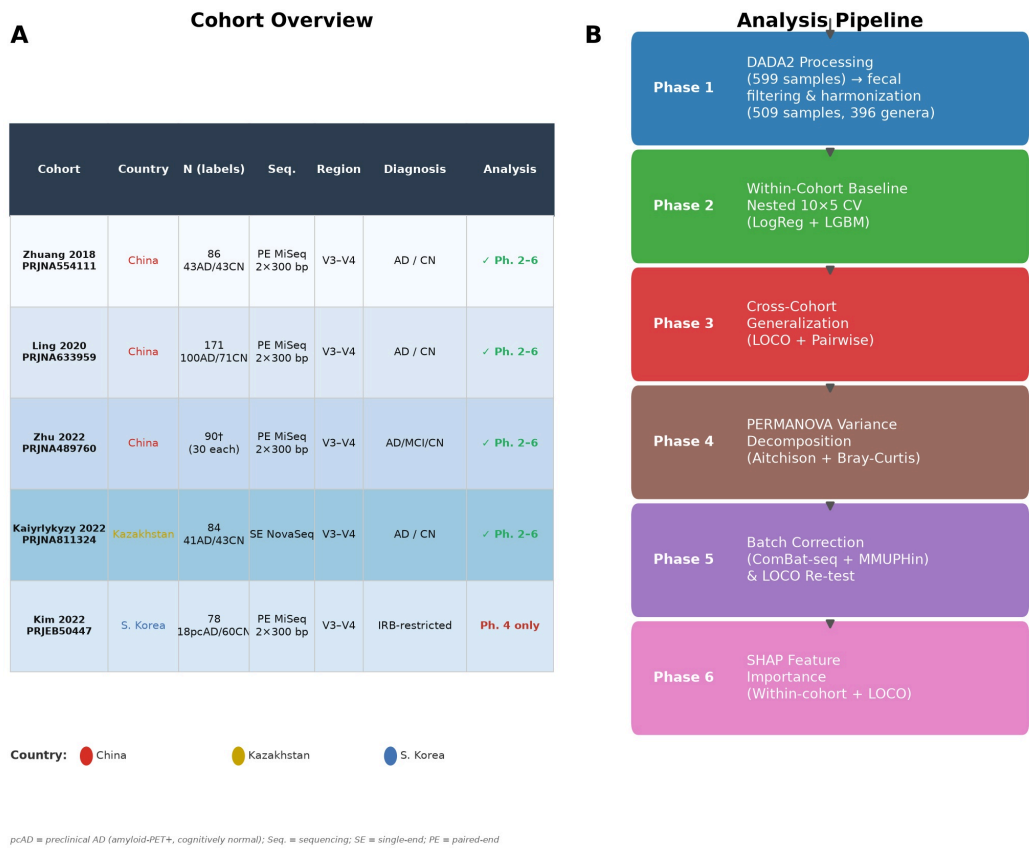


Figure 1

Figure 1. Study design. **(A)** Overview of the five 16S rRNA amplicon sequencing cohorts assembled for this study, including country of origin, sample size, sequencing mode, 16S variable region, diagnosis label availability, and analytic phases in which each cohort participates. **(B)** Summary of the six-phase analysis pipeline, from DADA2 processing through SHAP feature importance analysis.

Table 1. Cohort characteristics.

Cohort	Country	Accession	N (analyzed)	AD	CN	MCI	Sequencing	Library	Supervised Analysis
Zhuang 2018	China	PRJNA554111	86	43	43	—	MiSeq 2×300	Paired-end	Phases 2–6

Cohort	Country	Accession	N (analyzed)	AD	CN	MCI	Sequencing	Library	Supervised Analysis
Ling 2020	China	PRJNA633959	171	100	71	—	MiSeq 2×300	Paired-end	Phases 2–6
Zhu 2022†	China	PRJNA489760	90 (fecal)	30	30	30	MiSeq 2×300	Paired-end	Phases 2–6
Kaiyrlykyzy 2022	Kazakhstan	PRJNA811324	84	41	43	—	NovaSeq	Single-end	Phases 2–6
Kim 2022	South Korea	PRJEB50447	78	18*	60	—	MiSeq 2×300	Paired-end	Phase 4 only
Total			509	232	247	30			

Amyloid-PET positive preclinical AD; per-sample labels IRB-restricted and not publicly available. †Zhu 2022 deposited 180 samples (90 fecal F₋ + 90 blood B₋); only fecal samples retained for analysis.*

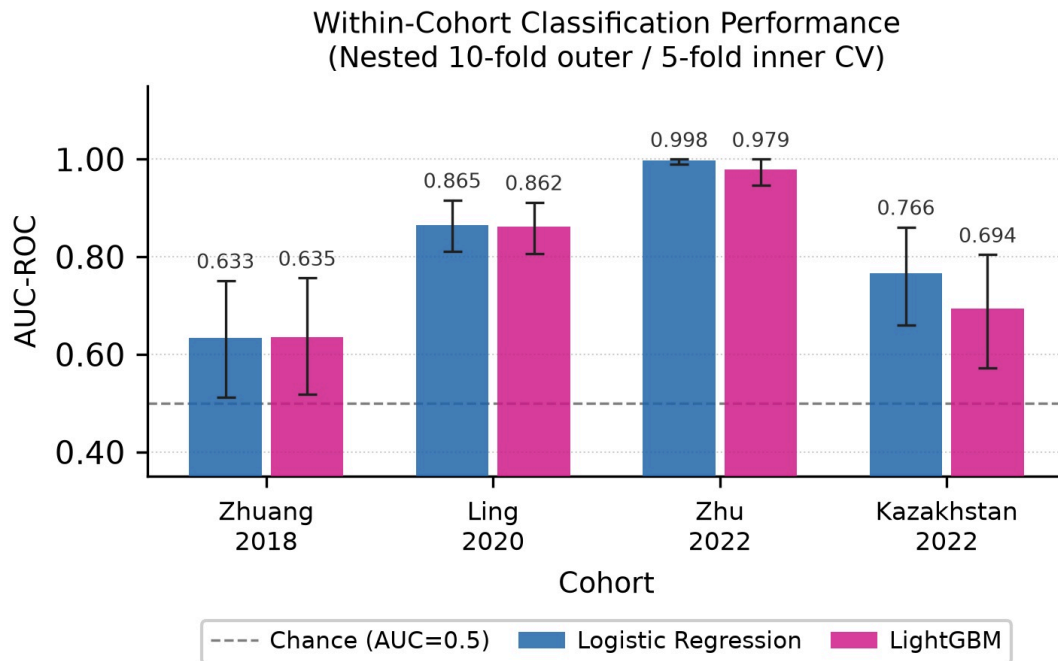


Figure 2

Figure 2. Within-cohort classification performance under nested cross-validation (10-fold outer / 5-fold inner CV). Bars show out-of-fold AUC-ROC; error bars represent 95% bootstrap confidence intervals (1,000 resamples). Dashed line = chance (AUC = 0.50). LogReg = L2-regularized logistic regression; LGBM = LightGBM.

Table 2. Within-cohort nested cross-validation performance (10-fold outer, 5-fold inner CV).

Cohort	Model	N	AUC-ROC	95% CI	Sensitivity	Specificity	Originally Reported AUC
Zhuang 2018	LogReg	86	0.633	[0.51–0.75]	0.814	0.512	~0.73 (Zhuang et al., 2018)
Zhuang 2018	LGBM	86	0.635	[0.52–0.75]	0.884	0.442	—

Cohort	Model	N	AUC-ROC	95% CI	Sensitivity	Specificity	Originally Reported AUC
Ling 2020	LogReg	171	0.865	[0.81–0.92]	0.780	0.859	~0.87 (Ling et al., 2021)
Ling 2020	LGBM	171	0.862	[0.80–0.91]	0.740	0.887	—
Zhu 2022	LogReg	60	0.998	[0.99–1.00]	0.967	1.000	~0.94 (Zhu et al., 2022)
Zhu 2022	LGBM	60	0.979	[0.94–1.00]	0.900	1.000	—
Kaiyrlykzy 2022	LogReg	84	0.766	[0.66–0.86]	0.829	0.605	~0.78 (Kaiyrlykzy et al., 2022)
Kaiyrlykzy 2022	LGBM	84	0.694	[0.57–0.80]	0.610	0.767	—

N=60 for Zhu 2022 binary analysis (30 AD + 30 CN fecal samples; MCI and blood samples excluded). Originally reported AUC values are approximate, derived from the primary publications using the best-performing classifier reported; direct numerical comparison is limited by differences in cohort subset, model type, and validation design. All 95% CIs use the diagnosis-stratified, participant-level bootstrap described in Section 2.5.2 (10,000 replicates), applied uniformly to every AUC CI in this paper.

3.3 Cross-Cohort Generalization Failure

LOCO cross-cohort AUC-ROC values revealed substantial and variable degradation in cross-cohort performance across all cohorts and model types (Figure 3A; Table 3). Mean LOCO AUC across the four cohorts was 0.640 for logistic regression and 0.647 for LightGBM—corresponding to mean drops of 0.176 and 0.146 AUC points respectively from within-cohort performance. One of eight model-cohort combinations produced LOCO AUC near chance: Zhuang 2018 with logistic regression (AUC=0.504 [0.38–0.63]).

The largest absolute logistic regression AUC drop was observed for Kazakhstan (within-cohort 0.766 → LOCO 0.564, $\Delta=0.202$). Zhu 2022 showed the largest LightGBM drop: within-cohort 0.979 → LOCO 0.758 ($\Delta=0.221$); its logistic regression drop was also substantial (0.998 → 0.813, $\Delta=0.185$). These results are consistent with LOCO models relying on predictive patterns learned from training cohorts that do not transfer fully to the held-out cohort, where cohort-specific compositional structure may contribute to within-cohort performance.

Ling 2020 also experienced substantial LOCO collapse: logistic regression 0.865 → 0.678 ($\Delta=0.187$); LightGBM 0.862 → 0.674 ($\Delta=0.188$).

Formal paired comparison of the within-cohort vs. LOCO AUC difference (Section 2.5.2). A paired, diagnosis-stratified, participant-level bootstrap (10,000 replicates, identical resampled participants used for both AUC values in each replicate) was used to formally compare the within-cohort and LOCO AUC difference for each of the eight cohort-model pairs, rather than relying on non-overlap of independently computed confidence intervals, which is not a formal test of a nonzero difference. Five of eight pairs showed a 95% bootstrap CI on the delta that excluded zero: Ling 2020 (logistic regression $\Delta=0.187$, 95% CI [0.10, 0.28]; LightGBM $\Delta=0.188$, 95% CI [0.10, 0.28]), Zhu 2022 (logistic regression $\Delta=0.184$, 95%

CI [0.07, 0.31]; LightGBM $\Delta=0.221$, 95% CI [0.11, 0.35]), and Kazakhstan logistic regression ($\Delta=0.201$, 95% CI [0.04, 0.36]). The remaining three pairs — Zhuang 2018 logistic regression ($\Delta=0.129$, 95% CI [-0.05, 0.30], proportion of bootstrap replicates with $\Delta \leq 0$: 0.079), Zhuang 2018 LightGBM ($\Delta=0.070$, 95% CI [-0.10, 0.24], proportion $\Delta \leq 0$: 0.214), and Kazakhstan LightGBM ($\Delta=0.104$, 95% CI [-0.05, 0.26], proportion $\Delta \leq 0$: 0.092) — showed the same-signed (within-cohort higher) point estimate but a 95% CI that included zero, and are not formally significant under this test despite the directional pattern being consistent across all eight pairs. We report this honestly rather than treating all eight LOCO drops as equally well-supported: the aggregate pattern of reduced cross-cohort performance is robust, but three of the eight individual cohort–model comparisons do not reach formal statistical support at this sample size (full results in Supplementary Table S10).

LOCO performance was heterogeneous across cohorts: Zhu 2022 achieved the highest LOCO AUC (logistic regression 0.813), while Zhuang 2018 collapsed to near chance. It should be noted that only one of the eight main model–cohort combinations fell to near chance, and five of eight exceeded 0.55; the results therefore demonstrate substantial and variable degradation rather than uniform failure.

Sensitivity analyses confirmed that generalization failure was not attributable to Kazakhstan, Zhuang 2018, or Ling 2020 individually (Supplementary Figure S4). When Kazakhstan was excluded and LOCO was performed on the three remaining Chinese cohorts, Zhuang 2018 remained near chance (LogReg 0.512, LightGBM 0.567), demonstrating that the generalization failure for Zhuang 2018 is not a Kazakhstan artifact. When Zhuang 2018 was excluded, LOCO on the remaining three cohorts still showed AUC values that were substantially below within-cohort baselines for Kazakhstan (LogReg 0.599, LightGBM 0.495). When Ling 2020 was excluded, Kazakhstan LOCO LogReg was 0.604. These sensitivity results confirm the robustness of the main finding across cohort compositions.

Training-only feature-selection sensitivity (Section 2.5.1) confirmed that this LOCO generalization-failure pattern is not an artifact of the global, all-cohort genus-selection step (Section 2.2). Re-deriving the retained genus set independently within each split — using only that split’s training-side samples, with the held-out cohort contributing zero information to feature selection — changed LOCO AUC by at most 0.066 in either direction (largest changes: Kazakhstan LightGBM $0.591 \rightarrow 0.525$, $\Delta=-0.066$; Kazakhstan LogReg $0.564 \rightarrow 0.609$, $\Delta=+0.045$; all other cohort–model combinations changed by ≤ 0.031). The retained genus count under strict training-only selection ranged from 150 (when Kazakhstan — a comparatively high-prevalence cohort for many genera — was excluded from feature selection during its own hold-out) to 373, versus 396 under the global rule, yet no cohort–model combination changed its qualitative classification (near-chance, intermediate, or robust). Within-cohort nested-CV AUC under training-only selection likewise moved by -0.083 to $+0.038$ (Supplementary Table S4). We conclude that the global feature-selection design in the main analysis, while a genuine methodological limitation in principle, did not materially inflate or deflate the magnitude of the reported cross-cohort generalization failure.

Table 3. LOCO cross-cohort AUC-ROC versus within-cohort nested CV baseline.

Test Cohort	Model	Within-Cohort AUC	LOCO AUC	LOCO 95% CI	AUC Drop
Zhuang 2018	LogReg	0.633	0.504	[0.38–0.63]	0.129
Zhuang 2018	LGBM	0.635	0.565	[0.44–0.69]	0.070
Ling 2020	LogReg	0.865	0.678	[0.59–0.75]	0.187
Ling 2020	LGBM	0.862	0.674	[0.59–0.75]	0.188

Test Cohort	Model	Within-Cohort AUC	LOCO AUC	LOCO 95% CI	AUC Drop
Zhu 2022	LogReg	0.998	0.813	[0.69–0.93]	0.185
Zhu 2022	LGBM	0.979	0.758	[0.63–0.87]	0.221
Kazakhstan	LogReg	0.766	0.564	[0.44–0.69]	0.202
Kazakhstan	LGBM	0.694	0.591	[0.47–0.71]	0.103
Mean	LogReg	0.816	0.640	—	0.176
Mean	LGBM	0.793	0.647	—	0.146

Paired-bootstrap formal significance for the AUC drop (Section 2.5.2; Supplementary Table S10): significant (95% CI on Δ excludes 0) for Ling 2020 (both models), Zhu 2022 (both models), and Kazakhstan LogReg; not significant at this sample size for Zhuang 2018 (both models) or Kazakhstan LGBM, despite all eight pairs showing the same-signed drop.

Pairwise single-cohort transfer AUC values (Figure 3B–C; Supplementary Table S1) ranged from 0.54 to 0.88 for logistic regression. The highest pairwise AUC was Ling 2020 $\rightarrow$ Zhu 2022 (logistic regression AUC=0.88), reflecting transfer between two large Chinese fecal cohorts with overlapping compositional structure. Kazakhstan $\rightarrow$ Ling 2020 also transferred reasonably (logistic regression AUC=0.74). Most other pairwise AUCs clustered between 0.54 and 0.70, consistent with the overall pattern of limited cross-cohort transferability.

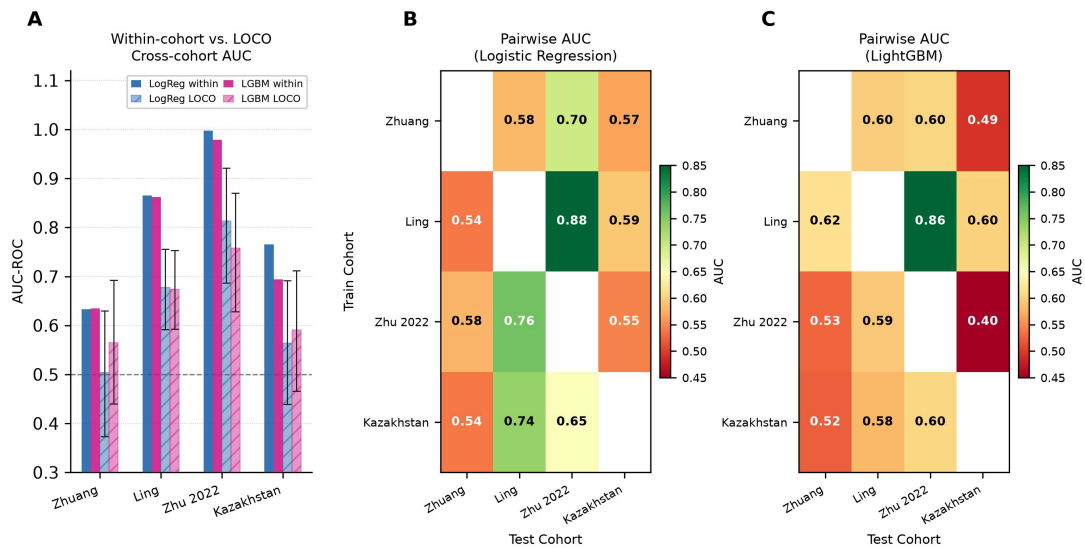


Figure 3

Figure 3. Cross-cohort generalization failure. (A) Within-cohort (solid bars) versus LOCO cross-cohort (hatched bars) AUC-ROC for logistic regression (blue) and LightGBM (pink). Error bars = 95% bootstrap CI. (B, C) Pairwise single-cohort-train / single-cohort-test AUC heatmaps for logistic regression and LightGBM respectively. Diagonal (within-cohort) values are omitted; colour scale: red = near chance, green = high AUC.

3.4 PERMANOVA Variance Decomposition

PERMANOVA on Aitchison distance across all five cohorts showed that cohort identity explained 19.3% of total compositional variance ($R^2=0.193$, $F=30.13$, $p=0.0001$, 9,999 permutations; Figure 4A). On Bray-Curtis dissimilarity, the cohort R^2 was 0.136 ($F=19.80$, $p=0.0001$). The cohort effect was substantially

larger than disease signal: in a marginal two-variable model restricted to the four labeled cohorts (n=401), cohort (marginal) explained $R^2=0.172$ ($F=28.03$, $p=0.0001$) versus diagnosis (marginal) $R^2=0.014$ ($F=6.82$, $p=0.0001$)—an approximately 12-fold disparity within the same model. This is an empirical statement about the relative magnitude of cohort- versus diagnosis-associated compositional heterogeneity in these data, not a decomposition of cohort into technical and biological components: “cohort” is a composite study-of-origin variable bundling sequencing platform, library preparation, recruitment site, diagnostic ascertainment, and unmeasured dietary/lifestyle factors, and PERMANOVA cannot separate these bundled sources (Section 2.6). It provides context for interpreting the batch-correction results in Section 3.5, but PERMANOVA marginal R^2 values do not by themselves establish that the cohort and diagnosis axes of variation overlap, nor do they explain why any particular batch-correction algorithm changed classifier performance in a specific direction; that relationship is discussed as a separate, more limited hypothesis in Section 4.2.

The ~12-fold disparity is sensitive to cohort composition and should not be read as a fixed, generalizable figure. Restricting the same marginal PERMANOVA specification to the three cohorts that are more technically comparable to one another — Zhuang 2018, Ling 2020, and Zhu 2022, all Chinese, paired-end MiSeq V3–V4 studies (Section 2.6) — reduced the disparity from approximately 12-fold to approximately 2.6-fold: cohort (marginal) $R^2=0.0588$ ($F=10.05$, $p=0.0001$) versus diagnosis (marginal) $R^2=0.0229$ ($F=7.83$, $p=0.0001$) in this n=317 subset (Supplementary Table S11). This subset is more technically comparable, not technically identical — recruitment site, exact diagnostic procedure, and unmeasured dietary factors still differ across these three studies — but removing Kazakhstan’s single-end/NovaSeq platform and Kim/KBASE’s separate country substantially changes the cohort-to-diagnosis variance ratio. We conclude that the ~12-fold figure computed on the full four-cohort set is not a stable, generalizable property of “cohort effects versus disease effects” in gut microbiome data; it is specific to the particular set of cohorts compared, and is driven substantially by the inclusion of cohorts that differ from one another in more than country of origin.

A logistic regression classifier trained to predict cohort identity from the CLR-transformed features across all 509 samples achieved macro-AUC=0.9917 (one-vs-rest, stratified 5-fold CV; per-cohort AUC: Kazakhstan 1.000, Kim 2022 0.997, Ling 2020 0.993, Zhuang 2018 0.989, Zhu 2022 0.980). Cohort membership is thus near-perfectly recoverable from the same features used for disease classification, confirming that cohort-specific information (whatever its technical or biological composition) dominates the compositional feature space.

Principal coordinates analysis (PCoA) on Aitchison distance (Figure 4B; n=509) visually confirmed strong cohort-level clustering, with partial overlap between Chinese cohorts (Zhuang 2018, Ling 2020) but clear separation of Kazakhstan and Zhu 2022. The Kim/KBASE 2022 cohort was clearly separated from all others along PC1, consistent with its preclinical amyloid-based staging, South Korean origin, and potentially distinct dietary patterns.

Beta-dispersion testing revealed significant heterogeneity in within-group spread across cohorts ($F=13.93$, $p=0.0001$, 9,999 permutations; Figure 4C), meaning that the PERMANOVA R^2 of 0.193 reflects both compositional centroid shifts and differences in within-cohort spread. Kazakhstan showed the greatest within-cohort variance, while the Chinese cohorts (Zhuang, Ling) were more tightly dispersed. Pairwise post-hoc betadisper comparisons (Section 2.6; Supplementary Table S12) show this overall dispersion difference is not uniform across all cohort pairs: Ling 2020 differs significantly in dispersion from every other cohort (all Holm-adjusted $p\leq 0.0063$), and is the single cohort driving most of the significant pairwise

comparisons, while shanghai2022 (Zhu 2022) vs. zhuang2018 (Holm $p=0.439$) and kazakhstan2022 vs. kbase2022 (Holm $p=0.439$) are not significantly different from one another in dispersion. Within the Chinese-MiSeq-only subset described above, dispersion remained significant ($F=7.946$, $p=0.0004$) and was again driven mainly by one cohort — zhuang2018 differed significantly from both ling2020 ($p=0.0001$) and shanghai2022 ($p=0.026$), while ling2020 and shanghai2022 did not differ significantly from each other ($p=0.343$). This dispersion heterogeneity is relevant to interpreting PERMANOVA effect sizes and is acknowledged in the Methods and Limitations sections.

Within-cohort PERMANOVA revealed highly variable disease-associated compositional effects: Ling 2020 ($R^2=0.053$, $F=9.53$, $p=0.0001$) and Zhu 2022 ($R^2=0.094$, $F=6.01$, $p=0.0001$) showed significant diagnosis effects, while Zhuang 2018 ($R^2=0.017$, $p=0.066$) and Kazakhstan ($R^2=0.022$, $p=0.069$) did not reach statistical significance at $\alpha=0.05$. These within-cohort disease effect sizes broadly rank-order with within-cohort AUC performance, providing a triangulating validation of the classifier results. The Zhu 2022 fecal-only $R^2=0.094$ represents the highest within-cohort disease effect among the four labeled cohorts, consistent with a detectable diagnosis-associated compositional difference in this dataset.

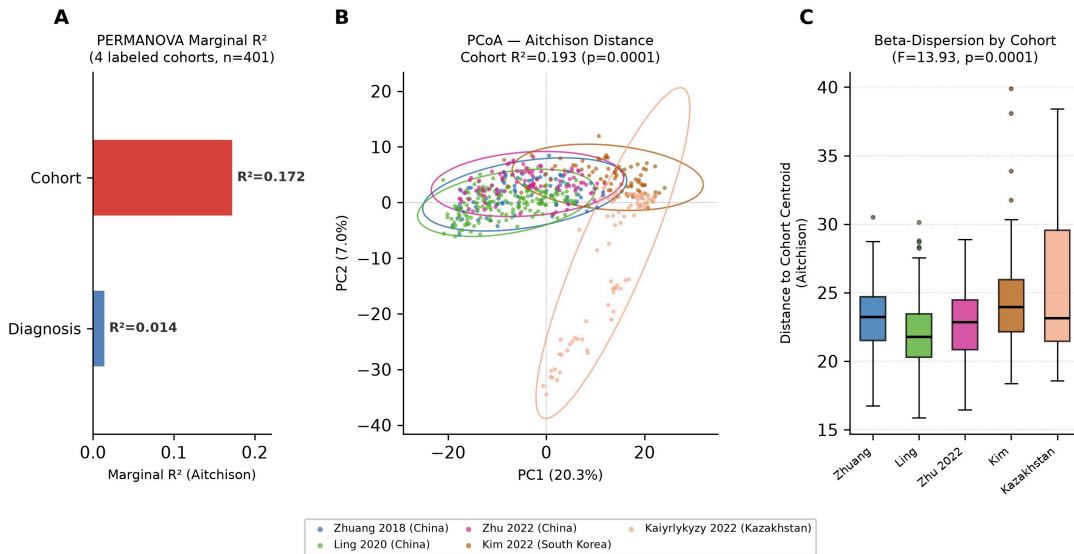


Figure 4

Figure 4. PERMANOVA variance decomposition and beta-diversity structure. (A) Marginal R^2 values from a two-variable PERMANOVA model on the four labeled cohorts ($n=401$): cohort $R^2=0.172$ and diagnosis $R^2=0.014$, shown as independent bars. These Type III marginal R^2 values are estimated independently and do not sum to 1. (B) PCoA of Aitchison distance matrix (all five cohorts; $n=509$). Cohort $R^2=0.193$ ($F=30.13$, $p=0.0001$, 9,999 permutations). Ellipses = 95% normal confidence ellipses. (C) Beta-dispersion: distance from each sample to its cohort centroid ($F=13.93$, $p=0.0001$, 9,999 permutations).

3.5 Label-Blind Transductive Batch Adjustment Does Not Improve Cross-Cohort Transferability

Batch correction was applied using ComBat-seq (on raw counts) and MMUPHin (on relative abundances) to the four labeled cohorts with cohort as the batch variable, under the two designs described in Section 2.7: a **label-blind** design (no diagnosis information used during correction; primary analysis) and a **label-informed** design (each sample's own true diagnosis label used during correction; exploratory sensitivity only) (Figure 5; Table 4).

Table 4. Mean LOCO AUC across four cohorts, by correction design and model.

Design	Method	LogReg Mean LOCO AUC	LogReg Mean Drop from Within-Cohort	LGBM Mean LOCO AUC	LGBM Mean Drop
—	Uncorrected	0.640	0.176	0.647	0.146
Label-blind (primary)	ComBat-seq	0.634	0.182	0.654	0.139
Label-blind (primary)	MMUPHin	0.639	0.176	0.648	0.145
Label-informed (exploratory)	ComBat-seq	0.493	0.322	0.555	0.238
Label-informed (exploratory)	MMUPHin	0.612	0.203	0.582	0.210

Label-blind correction (primary result): batch correction leaves cross-cohort generalization essentially unchanged. Under label-blind ComBat-seq and MMUPHin — the design that does not use any sample’s diagnosis label — mean LOCO AUC changed by no more than 0.007 in either direction relative to uncorrected, for either model (Table 4). No individual cohort collapsed: per-cohort logistic-regression LOCO AUC under label-blind ComBat-seq ranged from 0.525 (zhuang2018) to 0.769 (shanghai2022/Zhu 2022), closely tracking the uncorrected values (0.504–0.813) rather than diverging sharply from them. **Under label-blind transductive adjustment, which did not use diagnosis labels during correction, neither method materially changed mean LOCO AUC or recovered cross-cohort transferability under this design.**

Label-informed transductive sensitivity analysis (Supplementary Figure S5): severe degradation under this design was specific to the label-informed design itself. Under the label-informed design — which uses every sample’s own true diagnosis label during correction and therefore cannot represent a deployable or prospectively valid comparison — ComBat-seq reduced mean logistic-regression LOCO AUC from 0.640 to 0.493, with the Kazakhstan cohort collapsing to AUC=0.308 [0.20–0.42]. The same cohort under label-blind ComBat-seq scored 0.560, closely matching its uncorrected value (0.564). This side-by-side comparison indicates that the severe, cohort-specific collapse under the label-informed design was specific to that design — consistent with a correction model explicitly told each sample’s own diagnosis label (via `full_mod=TRUE/covariates="diagnosis"`) constructing a label-conditional transformation with no prospective analog — and therefore should not be interpreted as evidence that these correction methods generally degrade cross-cohort performance, or as evidence of an intrinsic incompatibility between ComBat-seq’s negative-binomial framework and 16S compositional data. MMUPHin’s label-informed effect was milder (LogReg mean: 0.640→0.612) but still did not improve on label-blind or uncorrected performance.

Within-cohort AUC on label-blind-corrected data was similar to uncorrected (within ± 0.05 for most cohort–method combinations; Supplementary Table S5), which provides a comparison free of diagnosis-label leakage during correction, because the label-blind correction never saw any sample’s diagnosis label. This comparison is not free of other confounding, however — it remains transductive (cohort membership and raw feature values are still used during correction) and retains whatever cohort/population confounding is otherwise present in the dataset. **We do not draw the same conclusion from within-cohort AUC on label-informed-corrected data — that comparison is additionally confounded by label use,** because the label-informed correction’s own covariate-preservation mechanism

(`full_mod=TRUE` for ComBat-seq; `covariates="diagnosis"` for MMUPHin) is explicitly designed to retain diagnosis-associated structure using every sample's own true label; any preserved within-cohort separability under that design cannot be attributed to correction-independent biological signal versus the correction algorithm's own label-preservation step. Concluding that “within-cohort AUC preservation rules out indiscriminate signal removal” requires a comparison free of diagnosis-label leakage during correction, which only the label-blind design provides; the label-blind within-cohort comparison supports this conclusion descriptively (Supplementary Table S5), without establishing it as strong biological evidence beyond the observed AUC values themselves, and we do not extend this conclusion to the label-informed results.

The empirical PERMANOVA finding that cohort explains approximately 12 times more compositional variance than disease status (marginal R^2 : 0.172 vs. 0.014; Section 3.4) provides context for why a correction method targeting cohort-associated variance might struggle to leave disease-associated variance fully intact, but it does not, by itself, establish that the two sources of variation overlap, nor does it explain the specific pattern observed here — namely that a properly label-blind correction left performance essentially flat while a label-informed one produced large, method- and cohort-specific movements. That specific pattern is more directly explained by the transductive, label-informed design itself (Section 2.7) than by the general cohort/diagnosis variance disparity, and we present it as such rather than as a demonstration that linear batch correction “cannot” separate cohort from disease signal in general.

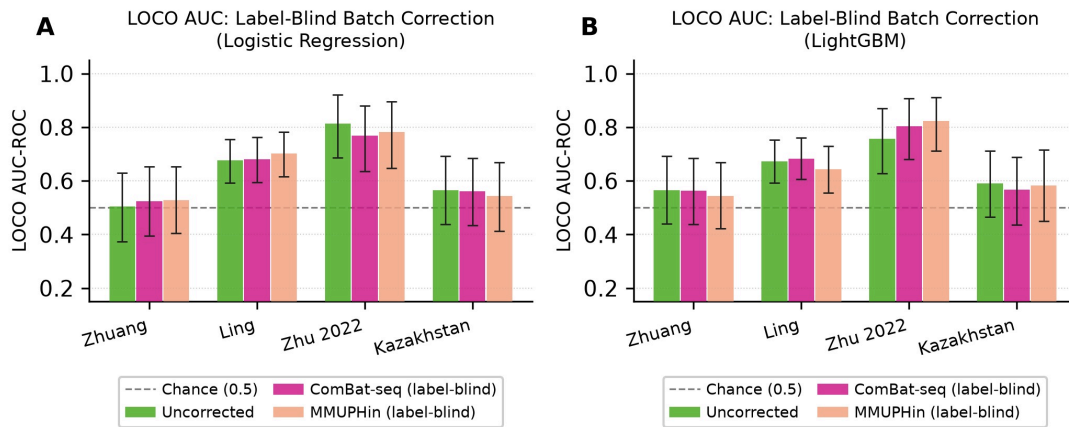


Figure 5

Figure 5. Effect of label-blind batch correction on LOCO cross-cohort AUC-ROC (primary analysis; no diagnosis information used during correction). (A) Logistic regression and (B) LightGBM LOCO AUC for each held-out cohort under three conditions: uncorrected (green), ComBat-seq label-blind corrected (magenta), and MMUPHin label-blind corrected (salmon). Error bars = 95% bootstrap CI. Dashed line = chance (0.50). Under this label-blind design, both methods left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), neither recovering nor substantially worsening cross-cohort generalization. An exploratory label-informed transductive sensitivity comparison, in which correction used each sample's own true diagnosis label and which produced substantially different, more severe degradation for some cohorts, is reported separately in Supplementary Figure S5 and is not treated as evidence about prospective performance (Section 2.7, Section 3.5).

3.6 SHAP Analysis Reveals Cohort-Specific Taxonomic Signatures and a Descriptive Coefficient-Stability Screen

To characterize the feature-level patterns underlying generalization failure, we computed out-of-fold SHAP values for within-cohort classifiers and LOCO-trained classifiers (Figure 6). Using an OOF design

avoids the in-sample overfitting bias identified in prior SHAP analyses of microbiome classifiers, where training-set SHAP values can reflect model memorization rather than the model's out-of-sample feature importance. Feature importance (mean |SHAP|) is reported for both models; direction is reported only for logistic regression, using the fitted-coefficient-based definition in Section 2.8 (not mean signed SHAP).

Minimal overlap in top predictive taxa across cohorts, assessed against two null models. Pairwise Jaccard similarity of the top-20 SHAP genera between cohort pairs averaged 0.135 for logistic regression and 0.058 for LightGBM (Figure 6C) — the mean of the six unique pairwise Jaccard values among the four labeled cohorts. Against the primary, shared-pool null distribution constructed to match this statistic exactly (100,000 replicates, each averaging the six pairwise Jaccards among four random top-20 sets drawn from the full 396-genus pool; Section 2.8), logistic regression overlap was significantly above chance (null mean 0.027 [95% CI: 0.009–0.049]; $p < 0.0001$), and LightGBM overlap was also significantly above chance at this N ($p = 0.0048$). A sensitivity check across $N \in \{10, 20, 50\}$ under this same shared-pool null (Supplementary Table S7) shows logistic regression significant at all three thresholds, while LightGBM was significant at $N = 20$ and $N = 50$ but not at $N = 10$ ($p = 0.45$).

Under a more realistic, per-cohort prevalence-restricted null, the LightGBM finding does not survive. The shared-pool null draws candidate genera as if every genus were equally available in every cohort, but per-cohort prevalence varies substantially (Section 2.8). Restricting each cohort's top-N ranking and random draws to genera reaching $\geq 20\%$ prevalence within that cohort's own AD/CN samples (eligible universe sizes: 131 for Zhuang 2018, 126 for Ling 2020, 115 for Zhu 2022, 356 for Kazakhstan; 100,000 replicates per model/N) changes the LightGBM result: LightGBM overlap was **not significant at any of $N = 10, 20$, or 50** under this prevalence-restricted null (observed mean Jaccard 0.018/0.063/0.167 vs. null mean 0.026/0.052/0.142; empirical $p = 0.80, 0.24, 0.061$ respectively; Supplementary Table S13). Logistic regression remained significant at all three thresholds under the prevalence-restricted null (observed 0.085/0.135/0.231 vs. null mean 0.026/0.052/0.142; $p = 0.0017, 0.00003, 0.00001$). We treat this as a genuine reversal rather than a discrepancy to be explained away: LightGBM's cross-cohort top-predictor overlap, which appeared to be a real if small signal under the shared-pool null, is not distinguishable from chance once the null model accounts for each cohort's own genus-prevalence structure. Logistic regression's overlap is robust to this change and remains the stronger and more reliable cross-cohort taxonomic-overlap finding of the two models.

Cohort-specific top predictors, and a concrete illustration of why coefficient-based direction was needed. The most discriminative genus for logistic regression was *Subdoligranulum* in Zhuang 2018, while *Akkermansia* was the top predictor in the remaining three cohorts (Zhu 2022 |SHAP|=0.228; Kazakhstan |SHAP|=0.136; Ling 2020 |SHAP|=0.270). Under the fitted model's coefficient, *Akkermansia* has a **stable positive coefficient in all four cohorts** (10 of 10 outer folds positive in every cohort; Supplementary Table S6) — higher CLR abundance consistently predicts higher AD log-odds under the fitted model, with no coefficient-based flip anywhere. This is a useful negative control for the direction-definition problem itself: the pooled mean SHAP value for this same genus is negative in Ling 2020 and Zhuang 2018 but positive in Zhu 2022 and Kazakhstan — an apparent sign difference that does **not** reflect the coefficient the model actually learned, and is instead a background-relative artifact of the kind described in Section 2.8. We report this discrepancy explicitly because it demonstrates, on real data from this study, why mean signed SHAP was not an adequate direction statistic and why the coefficient-based definition is used for every direction claim in this paper. For LightGBM, the top taxon per cohort was entirely distinct: *Lachnoclostridium* (Zhuang 2018), *Akkermansia* (Ling 2020), *Bacteroides* (Zhu 2022),

Castellaniella (Kazakhstan) (Figure 6A; Supplementary Figure S2); no direction is assigned to these LightGBM values (Section 2.8).

Coefficient-sign stability screen (logistic regression only). Using the pre-specified coefficient-stability criterion (Section 2.8: a stable positive coefficient sign in ≥ 8 of 10 outer folds in at least one cohort, and a stable negative coefficient sign in ≥ 8 of 10 outer folds in at least one other cohort, restricted to genera in the top-20 OOF mean-|SHAP| ranking in ≥ 2 cohorts), **eight genera met this descriptive stability criterion** (Figure 6B): *Agathobacter*, *Bifidobacterium*, *Coprococcus*, *Dorea*, *Lactobacillus*, *NK4A214* group, *Romboutsia*, and *Ruminococcus gnavus* group. We describe this deliberately as a descriptive screen rather than a statistically confirmed set of contradictions: the 10 outer-CV folds share substantially overlapping training sets (each fold's training data is 9/10 of the same cohort), so fold-level coefficient signs are not 10 independent replications, and no multiplicity-adjusted p-value is assigned to any individual taxon (Section 2.8, Limitations). This is a smaller and more conservative set than a less rigorous alternative approach that classifies any genus with opposite-signed pooled mean SHAP across two cohorts as a “flip” — an approach that, applied to this dataset, would identify twelve such genera; the stability requirement removes genera whose apparent flip is not reproducible across most of a cohort's own outer folds. **No comparable screen is applied to LightGBM:** a tree ensemble has no single coefficient-like, globally monotonic direction, and LightGBM's SHAP–feature relationships may be nonlinear or non-monotonic, so no scalar direction statistic — descriptive or otherwise — is computed or reported for it.

Romboutsia shows the cleanest cohort-level pattern among the eight taxa meeting the screen: a stable negative coefficient (higher relative abundance predicts lower AD log-odds under the fitted model) in all three Chinese cohorts and a stable positive coefficient in Kazakhstan (9/10 folds; Supplementary Table S6) — the one genus among the eight whose pattern cleanly separates by country in this dataset. We do not have a specific mechanistic hypothesis for *Romboutsia*.

Lactobacillus also meets the screen but its per-cohort pattern is messier: stable-negative in Zhuang 2018 only, and stable-positive in the other three cohorts (Ling 2020, Zhu 2022, Kazakhstan; Supplementary Table S6). Diet is one plausible unmeasured confounder for genus-level differences of this kind, but this dataset contains no dietary covariate data, and we do not attribute this or any other taxon's pattern to a specific mechanism.

Akkermansia does not meet the coefficient-stability screen at all: its fitted coefficient is stable and positive in all four cohorts (10/10 outer folds in every cohort; Supplementary Table S6), despite appearing to reverse sign under the pooled mean-SHAP statistic (above). This is the clearest illustration in this dataset of why coefficient-based, fold-stability-gated direction is necessary rather than a raw pooled-SHAP sign.

LOCO SHAP analysis: feature importance only, no cross-cohort direction assigned. When SHAP values were computed for LOCO-trained classifiers applied to held-out cohorts, *Akkermansia* was the top-importance predictor for two of four held-out cohorts — Zhuang 2018 (|SHAP|=0.470) and Zhu 2022 (|SHAP|=0.372) — while *Subdoligranulum* (|SHAP|=0.269) topped held-out Ling 2020 and *Christensenellaceae R-7* group (|SHAP|=0.368) topped held-out Kazakhstan. No genus ranked first across all four held-out conditions. Consistent with Section 2.8, we do not assign an AD/CN direction to these LOCO SHAP values: the LOCO SHAP background (the three training cohorts) and evaluation set (the held-out cohort) are drawn from systematically different populations by construction, so the sign of a LOCO mean SHAP value is confounded with the raw cross-cohort compositional difference quantified by PERMANOVA (Section 3.4) rather than reflecting a stable, model-derived direction. The absence of a

single top-importance genus across all four held-out cohorts is itself consistent with the classifier activating whichever genera most closely resemble its training-cohort signature rather than a disease marker that generalizes across populations.

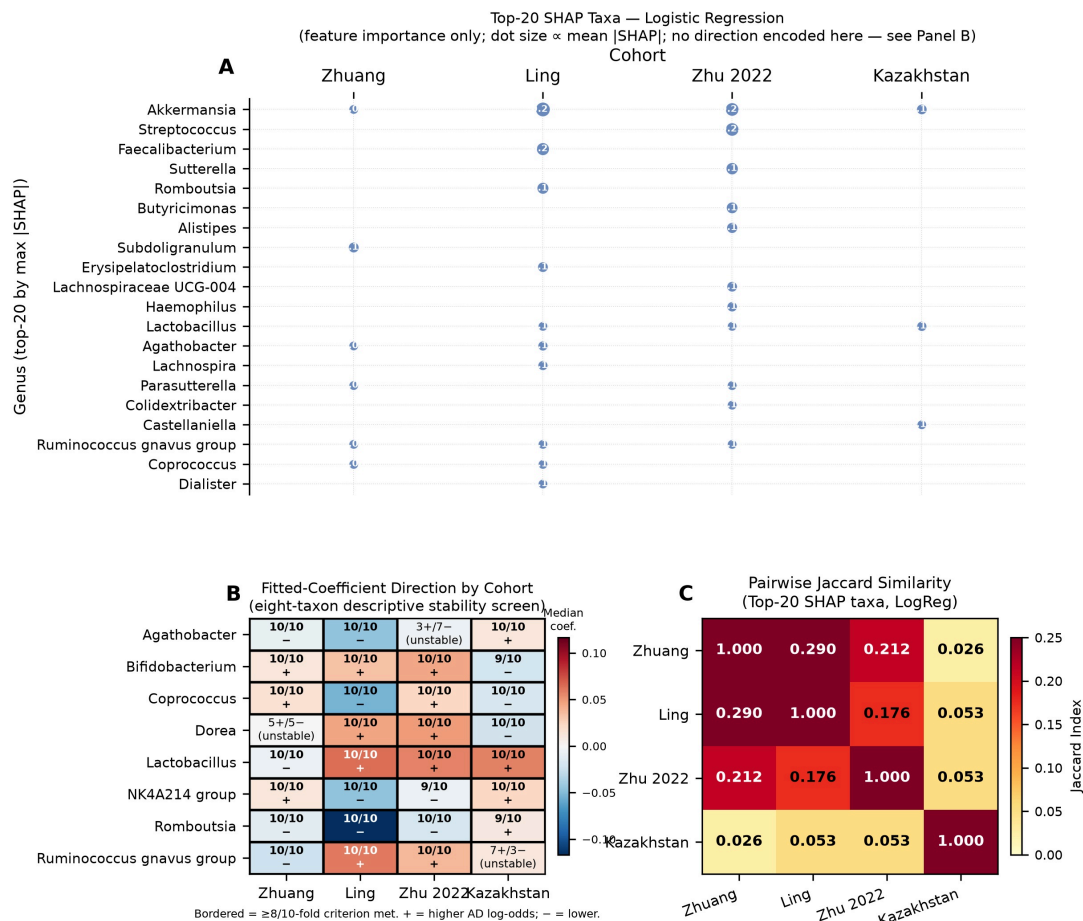


Figure 6

Figure 6. SHAP feature importance and coefficient-based direction analysis, logistic regression. **(A)** Top-20 SHAP genera by maximum |SHAP| across cohorts. Dot size $\propto$ mean |SHAP|; feature importance only — no direction is encoded in this panel. **(B)** Fitted-coefficient direction by cohort for the eight genera meeting a pre-specified descriptive coefficient-sign-stability screen (top-20 in ≥ 2 cohorts; a stable positive coefficient sign in $\geq 8/10$ outer folds in ≥ 1 cohort and a stable negative sign in $\geq 8/10$ folds in ≥ 1 other). Cell color/value = median fitted logistic-regression coefficient across the 10 outer folds (positive $\rightarrow$ higher predicted AD log-odds; negative $\rightarrow$ lower); cell text = the fold count backing that direction; bordered cells meet the $\geq 8/10$ stability criterion. This is a descriptive screen, not a multiplicity-corrected hypothesis test. **(C)** Pairwise Jaccard similarity of top-20 SHAP taxa between cohort pairs; mean of the six unique pairwise values = 0.135, significantly above a null matched to this statistic's construction (100,000 replicates of four random top-20 sets; $p < 0.0001$). A prevalence-restricted null sensitivity analysis (Supplementary Table S13) shows logistic regression remains significant while LightGBM does not.

4. Discussion

4.1 Systematic Quantification of Generalization Failure

This study provides a systematic, quantitative analysis of cross-cohort generalization limitations in gut microbiome-based AD classifiers. Across four labeled cohorts, two model types, two cross-cohort evaluation designs, and two batch correction methods, we consistently observe the same result: classifiers

trained within one cohort do not transfer reliably to another. The mean LOCO AUC drop of 0.176 points for logistic regression and 0.146 for LightGBM, with one of eight model-cohort combinations near chance, demonstrates that within-cohort performance figures — whether from the original cohort publications or from this study’s own nested-CV estimates — cannot be taken as evidence of a generalizable diagnostic signal. A formal paired bootstrap test (Section 3.3) supports this drop for five of the eight cohort-model pairs at conventional significance; the remaining three pairs show the same directional pattern without reaching formal significance at this sample size, and we report this distinction rather than treating all eight as equally well-supported.

This finding is consistent with the broader literature. The *Bacteroides* directionality reversal documented by Jemimah et al. (2023) across North American and Chinese cohorts, the contradictory reports for *Akkermansia* (Liu et al., 2020), and the highly cohort-specific patterns reported in individual papers all suggest population-level heterogeneity. A recent multi-cohort integration study (Wang et al., 2026, mSystems) similarly identified cross-geographic heterogeneity as a major obstacle; the present analysis complements that work by providing a systematic quantification of how severely within-cohort classifiers fail when applied to external populations, and by demonstrating that on average, neither method recovered cross-cohort transferability.

4.2 Batch-Adjustment Results and Limits of Mechanistic Interpretation

Two distinct findings must be kept separate here.

Empirical result: cohort identity explains substantially more compositional variance than disease status (Aitchison $R^2=0.193$ across all five cohorts; marginal $R^2=0.172$ cohort vs. 0.014 diagnosis in the four-cohort two-variable model — an approximately 12-fold disparity), and cohort membership is near-perfectly recoverable from the same CLR features (macro-AUC=0.9917). This 12-fold figure is not a fixed property of the data, however: restricted to the three more technically comparable Chinese paired-end MiSeq cohorts, the same marginal comparison falls to approximately 2.6-fold (cohort $R^2=0.059$ vs. diagnosis $R^2=0.023$; Section 3.4), and the significant beta-dispersion difference among cohorts (Section 3.4) means these R^2 values reflect both centroid separation and within-cohort spread, limiting a simple centroid-distance interpretation of “how separated” cohorts are. Taken together, compositional structure was strongly associated with cohort-of-origin in the evaluated datasets, but the exact magnitude of that association is sensitive to cohort composition and dispersion structure, not a single stable number. Separately and equally empirically: batch correction, evaluated under a label-blind design that did not use any sample’s diagnosis label during correction, left mean LOCO AUC essentially unchanged (within 0.007 AUC of uncorrected for both ComBat-seq and MMUPHin, both models; Section 3.5) — it neither recovered cross-cohort transferability nor substantially degraded it.

Hypothesis, clearly flagged as such: the much larger cohort-associated variance component provides plausible context for *why* correction targeting cohort-associated variance might struggle to isolate and remove only that variance without also touching disease-associated variance, if the two are not orthogonal in this compositional space. PERMANOVA marginal R^2 values do not establish that overlap directly, and we did not run an experiment (e.g., a simulation with a known, controlled degree of cohort/diagnosis-axis overlap) capable of testing it in this dataset. We therefore present it as a hypothesis consistent with the observed pattern, not as a demonstrated mechanism, and we no longer state that batch correction “cannot” work in general or that this analysis proves cohort and disease axes overlap.

Under the label-informed design, in which ComBat-seq and MMUPHin were both told every sample's own true diagnosis label during a single joint fit across all four labeled cohorts (Section 2.7), ComBat-seq's correction substantially worsened LOCO AUC (mean LogReg: 0.640→0.493, with one cohort collapsing to AUC=0.308). Repeating the identical comparison under a label-blind design — same cohorts, same raw features, same downstream LOCO classifiers, the only change being that no diagnosis label is supplied to the correction step — instead reproduces essentially flat LOCO AUC (Section 3.5, Table 4). This side-by-side comparison indicates that the severe degradation under the label-informed design was specific to that design and therefore should not be interpreted as evidence that these correction methods generally degrade cross-cohort performance, or as evidence of an intrinsic incompatibility between ComBat-seq's negative-binomial framework and 16S compositional data. One hypothesis consistent with this pattern, not established by it, is that a correction model explicitly told each sample's diagnosis label constructs a label-conditional transformation with no prospective analog. We therefore do not attribute the label-informed collapse to a property of ComBat-seq's modeling framework, and we retain the label-informed numbers only as an explicitly-labeled exploratory sensitivity illustrating what a label-aware, non-deployable correction can do (Section 3.5).

The finding that survives this correction — and that we consider the more robust and more useful result for the field — is narrower than originally stated: under the label-blind transductive evaluation, two widely used batch-correction methods designed for other data types (RNA-seq counts; general microbiome meta-analysis) did not recover cross-cohort transferability for AD classification in this dataset. This is a demonstration of ineffectiveness at the specific task of restoring cross-cohort generalization, not a demonstration that these methods actively destroy signal, nor an explanation of the underlying compositional-geometry reason such correction is difficult. The field may still need fundamentally different approaches — multi-cohort training with cohort as an explicit stratification variable, domain adaptation methods, or meta-analytic frameworks that explicitly model between-cohort heterogeneity rather than attempting to remove it — but we no longer claim to have shown mechanistically why standard correction fails, only that it does not succeed under the conditions tested here.

4.3 Implications of the Zhu 2022 Paired-Sample Design

The Zhu 2022 cohort deposited both blood microbiome (*B₊*) and fecal (*F₊*) samples from the same 90 participants (PRJNA489760). The blood samples were excluded from this analysis because blood and fecal microbiome communities represent fundamentally distinct compartments; including both would introduce a sample-type confound identical in structure to a batch effect, inflating within-cohort AUC through tissue-compartment discrimination rather than disease discrimination.

With fecal-only samples (n=60 binary), the near-perfect within-cohort AUC (logistic regression 0.998, LightGBM 0.979) is more difficult to attribute to a single confound. The small sample size (30 per class) is relevant: with 10-fold CV on 60 samples, each test fold contains approximately 6 samples, and single-fold AUC estimates can be highly variable. Zhu 2022 showed the largest detectable diagnosis-associated compositional effect ($R^2=0.094$, $F=6.01$, $p=0.0001$) of the four labeled cohorts. The repeated-CV and label-permutation analyses (Section 2.4.1, Section 3.2) show this near-perfect estimate is stable to fold assignment and clearly distinguishable from a chance-label null; despite this stability, the estimate may still overstate population-level performance because it is derived from only 60 participants and has not been independently replicated.

The LOCO result for Zhu 2022 (logistic regression AUC=0.813, the highest LOCO value among all cohorts) is informative: classifiers trained on the other three cohorts transfer to Zhu 2022 fecal samples better than to any other cohort. This suggests that the fecal disease signal in this cohort is somewhat consistent with signals in the Chinese and Kazakh training cohorts, unlike the spurious compartment-discrimination signal that would have characterized a hypothetical mixed-compartment analysis pooling fecal and blood samples together — a design this study avoided by excluding the blood samples at the outset (Section 2.4). We recommend that future AD microbiome studies report sample types explicitly and analyze each compartment independently before pooling.

4.4 Coefficient-Sign Stability Screen: Descriptive Cross-Cohort Direction Heterogeneity

Using a pre-specified coefficient-sign-stability criterion (≥ 8 of 10 outer folds agreeing in sign; Section 2.8), eight genera met this descriptive screen for logistic regression: *Agathobacter*, *Bifidobacterium*, *Coprococcus*, *Dorea*, *Lactobacillus*, *NK4A214 group*, *Romboutsia*, and *Ruminococcus gnavus group*. This is a smaller, more conservative set than the meta-analytic literature’s qualitative suggestion that some taxa carry contradictory directions of effect in different populations might imply, and we present it descriptively rather than as a statistically confirmed set of contradictions: the 10 outer-CV folds share substantially overlapping training data (each fold’s training set is 9/10 of the same cohort), so fold-level coefficient signs are not independent replications in the formal statistical sense, and we do not assign multiplicity-adjusted p-values to individual taxa (Section 2.8, Limitations). A less conservative alternative approach — classifying a flip as any genus with opposite-signed *pooled mean SHAP* across two cohorts (a background-relative quantity, not the model’s coefficient; Section 2.8) — would identify twelve such genera for logistic regression and, using the same statistic, a further twelve for LightGBM. Requiring the direction to be reproducible across most of a cohort’s own outer folds, and grounding it in the fitted coefficient rather than a background-dependent SHAP quantity, is why this paper reports eight genera for logistic regression and none for LightGBM: a tree ensemble has no single coefficient-like global direction to test for stability in the first place (Section 2.8), so no LightGBM directional-flip count is reported anywhere in this paper.

Romboutsia shows the cleanest cohort-level pattern among the eight: a stable negative fitted coefficient in all three Chinese cohorts, and a stable positive coefficient in Kazakhstan; we offer no mechanistic hypothesis for this genus. *Lactobacillus* also meets the screen, but its per-cohort pattern (stable-negative in Zhuang 2018 only; stable-positive in the other three cohorts) does not cleanly separate by country. Diet is a plausible unmeasured confounder for genus-level cohort differences generally, but this dataset has no dietary covariate data, and we do not attribute any specific taxon’s pattern to diet or any other mechanism without such data.

Akkermansia illustrates why the coefficient-based, stability-gated definition matters: its pooled mean SHAP value appears to reverse sign between cohorts (Section 3.6), but its fitted logistic-regression coefficient is positive and stable (10 of 10 outer folds) in all four cohorts (Supplementary Table S6), meaning the model never learned an inconsistent direction for this genus; the apparent sign reversal is an artifact of the background-relative SHAP statistic, not a property of the fitted model.

The critical implication of the eight taxa meeting this descriptive screen is that they may reflect biological population heterogeneity, technical study effects, or interactions between the two; the present design cannot disentangle these sources, nor does the screen itself establish that any individual taxon’s pattern exceeds what a non-independent, overlapping-fold coefficient estimate could show by chance. Standard batch correction methods, designed to remove technical batch effects, would not be expected to resolve

directional disagreements to the extent that these differences are driven by population-level biological differences rather than technical artifacts alone — a point that is, if anything, reinforced by Section 4.2’s finding that label-blind batch correction left cross-cohort AUC essentially unchanged rather than recovering it.

4.5 Implications for Clinical Translation

The results of this study suggest that gut microbiome-based AD classifiers are not ready for clinical deployment across heterogeneous external cohorts without prospective validation. A classifier trained on the Zhuang 2018 cohort and applied to the Kazakhstan cohort would perform at or below chance for logistic regression, and only modestly above chance even for LightGBM. These performance levels are clinically inadequate for any diagnostic application.

We propose three prerequisites for meaningful clinical translation of microbiome-based AD diagnostics: (1) **Multi-cohort harmonized training data**: classifiers should be trained on samples from multiple geographic and demographic populations simultaneously, with cohort or population as a stratification variable, rather than training within a single cohort and expecting transfer. (2) **Prospective cross-cohort validation as a publication standard**: any future study reporting a gut microbiome AD classifier should be required to validate performance on at least one independent cohort from a different geographic population before claiming clinical relevance. The within-cohort AUC, no matter how high, is an insufficient evidential standard. (3) **Covariate-adjusted models**: diet, geographic origin, antibiotic history, and other known microbiome-modulating covariates should be incorporated into classification models either as features or confounders, to reduce the influence of population-specific compositional baselines.

The multi-cohort harmonized collection approach has succeeded in other biomarker domains (e.g., the UK Biobank for genomic predictors, the ADNI consortium for neuroimaging). A parallel initiative—perhaps a consortium of AD microbiome studies committed to harmonized collection protocols and prospective cross-cohort sharing—would be the most direct route to a clinically actionable microbiome diagnostic.

4.6 Limitations

Several limitations of this study should be acknowledged explicitly.

Cohort search and selection: The search strategy documented in Section 2.1 and Supplementary Note S1 identified five publicly available cohorts meeting inclusion criteria. The cohort set is not exhaustive of all existing AD microbiome studies, and the inclusion of additional cohorts from non-Asian populations would strengthen the conclusions. The comparability of diagnostic criteria across cohorts (clinical AD diagnosis vs. amyloid-PET positivity) adds biological heterogeneity that cannot be fully disentangled from technical batch effects.

Mixed single-end and paired-end sequencing: Kazakhstan used single-end NovaSeq reads while all other cohorts used paired-end MiSeq 2×300 bp. Single-end reads yield lower taxonomic resolution at equivalent read length due to the absence of overlap-based error correction in DADA2, and different error models were fitted per cohort. This technical asymmetry contributes to between-cohort heterogeneity and cannot be disentangled from biological population differences in this dataset.

Kim/KBASE 2022 excluded from supervised analyses: Per-sample amyloid-PET diagnosis labels are IRB-restricted and not publicly deposited. The Kim 2022 cohort (78 participants; 18 amyloid-positive preclinical AD, 60 CN) therefore contributed only to PERMANOVA variance decomposition.

China overrepresentation: Three of the four labeled cohorts are Chinese. The cross-cohort experiment is therefore primarily a China-internal transfer (Zhuang ↔ Ling ↔ Zhu 2022) plus transfer to/from a single Central Asian cohort. A more geographically balanced design—incorporating North American, European, and African populations—would be needed to generalize the conclusions to the full range of human diversity.

Partial SRA deposit for Zhu 2022: The public archive contains 180 samples representing 90 participants, a subset of the 302 participants analyzed in the published study (PRJNA489760). The deposited subset may not be a random sample of the full study population, potentially introducing selection bias of unknown direction.

Global genus-selection feature universe: The 396-genus feature set (Section 2.2) was selected using $\geq 20\%$ prevalence pooled across all five cohorts simultaneously, meaning the eventual held-out cohort in every LOCO/pairwise experiment contributed prevalence information to the feature universe. A dedicated training-only sensitivity analysis (Section 2.5.1, Section 3.3), run once under the same prespecified `random_state=42` cross-validation partition used throughout this paper, re-derived genus selection independently within every split and found LOCO AUC changed by at most 0.066 — the qualitative generalization-failure conclusion is not an artifact of this design — but the global selection remains a methodological limitation in principle, and we recommend training-only feature selection as the default design for future cross-cohort microbiome classifier studies rather than a post hoc sensitivity check. Repeated cross-validation with multiple random seeds was not performed for this sensitivity analysis, so the 0.066 figure should be read as a single-partition estimate rather than one adjusted for ordinary fold-assignment variability.

PERMANOVA cannot decompose cohort into technical and biological sources, and the cohort-versus-diagnosis variance disparity is sensitive to which cohorts are compared. “Cohort” in this study is a composite study-of-origin variable bundling sequencing platform, library preparation, recruitment site, diagnostic ascertainment, and unmeasured dietary/lifestyle differences; PERMANOVA and betadisper estimate the total variance associated with this composite variable, not a technical-batch-effect estimate specifically, and no method used in this paper separates the two (Section 2.6). The approximately 12-fold cohort-versus-diagnosis disparity computed across all four labeled cohorts fell to approximately 2.6-fold when restricted to the three more technically comparable Chinese paired-end MiSeq cohorts (Section 3.4), indicating that this ratio is not a fixed property of “cohort effects” in gut microbiome data generally but depends substantially on which cohorts happen to be compared. Readers should not extrapolate the ~12-fold figure to other cohort sets.

Jaccard overlap significance depends on the null model chosen, and one finding reverses depending on that choice. LightGBM’s cross-cohort top-predictor overlap was significant at $N=20/50$ (not $N=10$) under a null that draws candidate genera uniformly from the shared 396-genus pool, but was not significant at any tested N under a null restricted to each cohort’s own $\geq 20\%$ -prevalence genera (Section 3.6). We report both nulls and the resulting reversal rather than selecting whichever null supports a stronger claim. Logistic regression’s overlap was significant under both null designs at all tested N and is the more robust of the two cross-model findings.

Batch correction: two designs, both transductive, neither prospective. As detailed in Section 2.7, both the label-blind (primary) and label-informed (exploratory) batch-correction designs are transductive — both estimate batch-effect parameters using the eventual held-out cohort’s own feature values and cohort

membership — and neither should be extrapolated to prospective deployment. The label-informed design additionally uses every sample's own true diagnosis label during correction and is retained only as an exploratory sensitivity analysis; its results (including the severe Kazakhstan collapse under label-informed ComBat-seq, Section 3.5) should not be read as evidence about batch correction's effect on biological signal in general, since they are substantially confounded by the label-informed design itself (Section 4.2).

Repeated cross-validation is a fold-stability check, not independent replication; three of eight within-cohort-vs-LOCO comparisons are not formally significant. The 50-repetition repeated nested-CV analysis for Zhu 2022 (Section 2.4.1, Section 3.2) uses the same 60 participants in every repetition and therefore quantifies sensitivity to fold assignment, not sampling variability across independent samples; the label-permutation test in the same section is the analysis that formally tests whether the observed AUC exceeds chance. Separately, the formal paired bootstrap test of within-cohort-vs-LOCO AUC differences (Section 2.5.2, Section 3.3) found three of eight cohort–model pairs (Zhuang 2018, both models; Kazakhstan LightGBM) did not reach a 95% CI excluding zero, despite all eight showing the same-signed drop; we report this honestly rather than treating the aggregate mean drop as equally well-supported for every individual cohort–model pair.

Cross-sectional design and observational data: All cohorts used cross-sectional sampling. Causal inference about microbiome–AD relationships is not supported by this design. All language throughout this paper refers to associations and predictive performance, not causal effects.

SHAP and coefficient-direction interpretability limitations: For logistic regression, direction is now defined from the fitted model's coefficient sign (Section 2.8), not from mean signed SHAP; a stable coefficient sign is a statement about what the fitted model consistently learned across cross-validation folds, conditional on regularization (L2, not elastic-net — see below) and collinearity among CLR features, and is not a direct estimate of biological association, its statistical significance, or causal direction. **The $\geq 8/10$ -fold coefficient-sign-stability screen is descriptive, not a formal hypothesis test.** The 10 outer-CV folds within a cohort are fit on substantially overlapping training sets (each omits only 1/10 of that cohort's samples), so they are not independent statistical replications; we do not apply a multiplicity correction across the candidate taxa screened, and no p-value is assigned to any individual taxon's stability count. The eight genera meeting this screen should be treated as empirical patterns worthy of targeted, covariate-adjusted, and independently-replicated follow-up, not as statistically confirmed mechanistic findings, and no dietary or other mechanistic attribution is made for any of them in the absence of covariate data. For LightGBM, no coefficient-like global direction exists, and no directional screen or flip count of any kind is reported for it anywhere in this paper. All LOCO SHAP values (either model) are reported as feature-importance rankings only, without any assigned cross-cohort direction, because the LOCO SHAP background (training cohorts) and evaluation set (held-out cohort) come from systematically different populations, confounding sign with raw cross-cohort compositional differences.

Logistic regression models use L2 (ridge), not elastic-net, regularization: The logistic-regression models throughout this paper (`solver="saga"`) are configured with `penalty="l2"` (not `"elasticnet"`), and are therefore L2 (ridge)-regularized. An `l1_ratio` parameter is present in the estimator configuration but is silently ignored by scikit-learn under the L2 penalty. One consequence is that essentially no fitted coefficient is exactly zero in the coefficient-stability analysis (Section 2.8), since L2 regularization does not perform the sparse variable selection that elastic-net or L1 penalties would.

5. Conclusion

We conducted a systematic cross-cohort generalization evaluation of gut microbiome-based Alzheimer's disease classifiers, spanning five independent cohorts from three countries. Within-cohort classifiers achieved AUC-ROC values of 0.633–0.998, consistent with published reports; the near-perfect Zhu 2022 result was stable across 50 repeated cross-validation partitions (mean AUC 0.9955–0.9760 depending on model) and was distinguishable from a label-permutation null. LOCO cross-cohort AUC values fell to 0.504–0.813, a mean drop of 0.176 AUC points for logistic regression and 0.146 for LightGBM; a formal paired bootstrap test found this drop statistically supported for five of eight cohort–model pairs (Ling 2020 and Zhu 2022, both models; Kazakhstan logistic regression), with the remaining three pairs (Zhuang 2018, both models; Kazakhstan LightGBM) showing the same directional pattern without reaching formal statistical support at this sample size. This generalization-limitation pattern was robust to a strict training-only re-derivation of the feature-selection step (LOCO AUC changed by at most 0.066 when held-out-cohort information was fully excluded from genus selection). Batch correction was evaluated under two designs: a label-blind design (no diagnosis information used) left mean LOCO AUC essentially unchanged relative to uncorrected (within 0.007 AUC for both methods and models), and did not recover cross-cohort transferability; an exploratory label-informed design (using each sample's own true diagnosis label, which cannot estimate prospective performance) produced large, cohort-specific degradation attributable in substantial part to the label-informed design itself rather than to an intrinsic property of the correction algorithms. PERMANOVA revealed that cohort identity explained substantially more compositional variance than disease status (marginal R^2 : 0.172 vs. 0.014, approximately 12-fold, across the four labeled cohorts) — an empirical measure of relative heterogeneity in a composite, study-of-origin variable that PERMANOVA cannot decompose into technical and biological sources, and that does not by itself explain the batch-correction results. This disparity is sensitive to cohort composition: restricted to the three more technically comparable Chinese MiSeq cohorts, it falls to approximately 2.6-fold ($R^2=0.059$ vs. 0.023), and pairwise dispersion comparisons show the overall dispersion effect is driven predominantly by one cohort rather than uniformly. A cohort identity classifier achieved macro-AUC=0.9917, confirming that cohort structure dominates the feature space regardless of its technical or biological composition. Sensitivity analyses confirmed that generalization failure persists across all three prespecified cohort-exclusion configurations. Out-of-fold SHAP analysis identified near-zero taxon overlap between cohort-specific predictive signatures (mean Jaccard 0.135 for logistic regression and 0.058 for LightGBM); against a null matched to the observed statistic's construction, logistic regression overlap was significant at all tested N, while LightGBM was significant at N=20/50 but not N=10 under a shared-genus-pool null, and was not significant at any N under a more realistic per-cohort prevalence-restricted null — a result we report as a genuine reversal rather than preserving the earlier, less rigorous finding. Using a pre-specified descriptive coefficient-sign-stability screen (not a formal hypothesis test), eight genera for logistic regression showed opposite, stable fitted-coefficient direction across cohorts; no comparable screen is applied to LightGBM, which has no single coefficient-like global direction. These findings demonstrate substantial cross-cohort generalization limitations among the evaluated publicly available 16S datasets; the two batch-correction methods tested under the label-blind transductive design did not recover transferability. Multi-cohort harmonized prospective studies and prospective cross-cohort validation are needed before microbiome-based AD biomarkers can achieve clinical utility.

Data Availability Statement

All raw sequencing data analyzed in this study are publicly available from NCBI SRA or ENA under accession numbers PRJNA554111 (Zhuang 2018), PRJNA633959 (Ling 2020), PRJNA489760 (Zhu 2022), PRJNA811324 (Kaiyrylkyzy 2022), and PRJEB50447 (Kim 2022). Analysis code, processed data tables, and figure generation scripts are available at <https://github.com/carhanc/microbiome-ad-generalization>.

Author Contributions

A.C. conceived the study, developed and executed the complete analysis pipeline, interpreted the results, and wrote the manuscript.

Funding

The author declares that no external funding was received for the conduct of this research.

Acknowledgements

The author thanks the original study teams of all five cohorts for depositing their data in publicly accessible archives, without which this analysis would not have been possible.

Conflict of Interest

The author declares no conflict of interest.

References

- Callahan, B.J., McMurdie, P.J., Rosen, M.J., Han, A.W., Johnson, A.J.A., Holmes, S.P. (2016). DADA2: High-resolution sample inference from Illumina amplicon data. *Nature Methods*, 13(7), 581–583. <https://doi.org/10.1038/nmeth.3869>
- Cryan, J.F., O’Riordan, K.J., Cowan, C.S.M., Sandhu, K.V., Bastiaanssen, T.F.S., Boehme, M., et al. (2019). The microbiota–gut–brain axis. *Physiological Reviews*, 99(4), 1877–2013. <https://doi.org/10.1152/physrev.00018.2018>
- Davison, A.C., Hinkley, D.V. (1997). *Bootstrap Methods and Their Application*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511802843>
- Harach, T., Marungruang, N., Duthilleul, N., Cheatham, V., Mc Coy, K.D., Frisoni, G., et al. (2017). Reduction of Abeta amyloid pathology in APPPS1 transgenic mice in the absence of gut microbiota. *Scientific Reports*, 7(1), 41802. <https://doi.org/10.1038/srep41802>

- Jemimah, S., Chabib, C.M.M., Hadjileontiadis, L., AlShehhi, A. (2023). Gut microbiome dysbiosis in Alzheimer's disease and mild cognitive impairment: a systematic review and meta-analysis. *PLOS ONE*, 18(5), e0285346. <https://doi.org/10.1371/journal.pone.0285346>
- Jung, J.H., Kim, G., Byun, M.S., Lee, J.H., Yi, D., Park, H., Lee, D.Y., KBASE Research Group. (2022). Gut microbiome alterations in preclinical Alzheimer's disease. *PLOS ONE*, 17(12), e0278276. <https://doi.org/10.1371/journal.pone.0278276>
- Kaiyrlykyzy, A., Kozhakhmetov, S., Babenko, D., Zholdasbekova, G., Alzhanova, D., Olzhayev, F., Baibulatova, A., Kushugulova, A.R., Askarova, S. (2022). Study of gut microbiota alterations in Alzheimer's dementia patients from Kazakhstan. *Scientific Reports*, 12, 15115. <https://doi.org/10.1038/s41598-022-19393-0>
- Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., et al. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems* 30, pp. 3146–3154.
- Kowalski, K., Mulak, A. (2019). Brain-gut-microbiota axis in Alzheimer's disease. *Journal of Neurogastroenterology and Motility*, 25(1), 48–60. <https://doi.org/10.5056/jnm18087>
- Ling, Z., Zhu, M., Liu, X., Shao, L., Cheng, Y., Yan, X., et al. (2021). Structural and functional dysbiosis of fecal microbiota in Chinese patients with Alzheimer's disease. *Frontiers in Cell and Developmental Biology*, 8, 634069. <https://doi.org/10.3389/fcell.2020.634069>
- Liu, P., Wu, L., Peng, G., Han, Y., Tang, R., Ge, J., et al. (2019). Altered microbiomes distinguish Alzheimer's disease from amnesic mild cognitive impairment and health in a Chinese cohort. *Brain, Behavior, and Immunity*, 80, 633–643. <https://doi.org/10.1016/j.bbi.2019.05.008>
- Liu, S., Gao, J., Zhu, M., Liu, K., Zhang, H.L. (2020). Gut microbiota and dysbiosis in Alzheimer's disease: implications for pathogenesis and treatment. *Molecular Neurobiology*, 57(12), 5026–5043. <https://doi.org/10.1007/s12035-020-02073-3>
- Lundberg, S.M., Lee, S.I. (2017). A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems* 30, pp. 4765–4774.
- Ma, S., Shungin, D., Mallick, H., Schirmer, M., Nguyen, L.H., Kolde, R., Franzosa, E., Vlamakis, H., Xavier, R., Huttenhower, C. (2022). Population structure discovery in meta-analyzed microbial communities and inflammatory bowel disease using MMUPHin. *Genome Biology*, 23(1), 208. <https://doi.org/10.1186/s13059-022-02753-4>
- Martín-Fernández, J.A., Barceló-Vidal, C., Pawlowsky-Glahn, V. (2003). Dealing with zeros and missing values in compositional data sets using nonparametric imputation. *Mathematical Geology*, 35(3), 253–278. <https://doi.org/10.1023/A:1023866030544>
- Oksanen, J., Simpson, G.L., Blanchet, F.G., Kindt, R., Legendre, P., Minchin, P.R., et al. (2024). *vegan: Community Ecology Package*. R package version 2.6-8. <https://CRAN.R-project.org/package=vegan>
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
- Plovier, H., Everard, A., Druart, C., Depommier, C., Van Hul, M., Geurts, L., et al. (2017). A purified membrane protein from *Akkermansia muciniphila* or the pasteurized bacterium improves metabolism in obese and diabetic mice. *Nature Medicine*, 23(1), 107–113. <https://doi.org/10.1038/nm.4236>

- Vogt, N.M., Kerby, R.L., Dill-McFarland, K.A., Harding, S.J., Merluzzi, A.P., Johnson, S.C., et al. (2017). Gut microbiome alterations in Alzheimer's disease. *Scientific Reports*, 7(1), 13537. <https://doi.org/10.1038/s41598-017-13601-y>
- Wang, J., Liu, H., Lai, H., Bao, Y., Shen, M., Li, C., Ma, L., Wu, T., Yang, S., Du, X., O'Brien, T.J., Zhang, J., Zhang, L. (2026). Identifying candidate gut microbiota indicators for Alzheimer's disease through integrated data. *mSystems*, 11(4). <https://doi.org/10.1128/msystems.01754-25>
- World Health Organization. (2023). *Dementia*. WHO Fact Sheet. <https://www.who.int/news-room/fact-sheets/detail/dementia>
- Zhang, Y., Parmigiani, G., Johnson, W.E. (2020). ComBat-seq: Batch effect adjustment for RNA-seq count data. *NAR Genomics and Bioinformatics*, 2(3), lqaa078. <https://doi.org/10.1093/nargab/lqaa078>
- Zhu, Z., Ma, X., Wu, J., Xiao, Z., Wu, W., Ding, S., Zheng, L., Liang, X., Luo, J., Ding, D., Zhao, Q. (2022). Altered gut microbiota and its clinical relevance in mild cognitive impairment and Alzheimer's disease: Shanghai Aging Study and Shanghai Memory Study. *Nutrients*, 14(19), 3959. <https://doi.org/10.3390/nu14193959>
- Zhuang, Z.Q., Shen, L.L., Li, W.W., Fu, X., Zeng, F., Gui, L., et al. (2018). Gut microbiota is altered in patients with Alzheimer's disease. *Journal of Alzheimer's Disease*, 63(4), 1337–1346. <https://doi.org/10.3233/JAD-180176>
-